

Dynamics of CEO Disclosure Style

Khrystyna Bochkay
Roman Chychyla
Dhananjay (DJ) Nanda
University of Miami

ABSTRACT: We examine changes in CEOs' disclosure styles in quarterly earnings conference calls over their tenure. Our longitudinal analysis of newly hired CEOs shows that CEOs' forward-looking disclosures and their disclosures' relative optimism decline in their tenure. Further, externally hired and inexperienced CEOs are more future-oriented, and younger CEOs exhibit greater optimism in their disclosures. We also find that non-CEO executives' disclosure styles remain time-invariant over their CEOs' tenure. Our evidence is consistent with uncertainty reduction about managers' ability over their tenure (1) reducing the demand for and the supply of forward-looking disclosures, and (2) attenuating managerial career concerns leading to the decline in disclosure optimism.

JEL Classifications: D22; D70; D82; D83; L20; M12.

Keywords: CEO tenure; ability uncertainty; career concerns; voluntary disclosure; forward-looking information; disclosure tone; earnings conference calls.

I. INTRODUCTION

Researchers have long examined how firms' disclosures create economic benefits by resolving stakeholders' uncertainty regarding their firms' performance and prospects. Information disclosures increase a firm's ability to access capital by potentially reducing outside parties' (investors and creditors) perceived uncertainty about their claims (Diamond and Verrecchia 1991; Leuz and Verrecchia 2000). The central premise of this inquiry is that uncertainty about a firm's future, as well as its manager's private information about plans and strategies, affects outsiders' demand for information. Appointment of a new CEO exacerbates outsiders' uncertainty, as the "match" between a firm's business model and its CEO's ability is unknown and can only be ascertained after observing performance and periodic reports, i.e., over time.¹ Management ability uncertainty increases the demand for information while also exacerbating managerial career concerns, as labor market participants revise their beliefs about managerial ability based on available information. Since both the uncertainty about CEOs' ability to enhance firm value and their managerial career concerns change with employment experience (Pan, Wang, and Weisbach 2015), we empirically examine the dynamics of CEO disclosure styles in quarterly earnings conference calls over their tenure as firm leaders. Using a longitudinal research design that examines newly hired CEOs' disclosure styles over the course of their tenure, and controlling for firm performance, other characteristics, and firm-CEO fixed effects, we find that both the amount of forward-looking disclosures made by CEOs during quarterly earnings conference calls, as well as their overall (forward- and non-forward-looking) disclosures' relative optimism, decline over their tenure.

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¹ Consistent with Hermalin and Weisbach (1998), we use the term "ability" to parsimoniously represent actions, plans, strategies, and knowledge.

CEOs' disclosures to their teams, employees, and outside constituencies is an essential part of their duties as their organizations' leaders.² Since CEOs play a preeminent role in affecting their firm policies, empirical studies have linked firm and executive characteristics to their disclosures and other managerial decisions. Studies have examined whether firm performance explains variation in managers' disclosure choices (Li 2008, 2010a), how firms' proprietary costs and litigation environment relate to management disclosure (Demers and Vega 2008; Rogers, Van Buskirk, and Zechman 2011), how these disclosures change relative to changes in firm performance and operations (Brown and Tucker 2011), and whether managers' personal characteristics, such as education, professional certification, and demographic background, explain variations in their disclosures and operating policies (Bertrand and Schoar 2003; Davis, Ge, Matsumoto, and Zhang 2015). Recent work by Li, Minnis, Nagar, and Rajan (2014) shows that executive participation in earnings conference calls is positively related to their knowledge and expertise about business operations. These studies significantly contribute to our understanding of executive disclosures; however, they are cross-sectional in design and do not examine potential temporal variations due to uncertainty resolution affecting the demand for and the supply of information. Our work supplements this literature by documenting the dynamics of CEOs' disclosure styles and identifies discernible temporal variation in managers' forward-looking disclosures and optimism over their tenure as firm leaders.

Our study uses a sample of 573 unique individuals that begin their appointments as CEOs of publicly listed U.S. firms after 2005 and for whom we can obtain transcripts of quarterly earnings conference calls in which they participate. Our primary reason for using earnings conference calls is that we are able to capture CEO-specific disclosures since we can identify statements made by each call participant (CEOs, other firm officers, and participating analysts and investors). This enables us to construct our disclosure measures specifically for the CEO, as well as for other executives (non-CEO officers) who participate in the conference call.³

Recent work has shown that conference calls are an important medium through which firms publicly disclose information (Frankel, Johnson, and Skinner 1999; Matsumoto, Pronk, and Roelofsens 2011). In contrast to mandatory filings, such as 10-K and 10-Q reports, that are more formal and often repetitive, earnings conference calls are public spoken events that connect management with investors and analysts, and allow learning about management plans, strategies, and performance on an ongoing basis.⁴ Finally, earnings calls have (mostly) a fixed duration, which results in little (if any) variation over time (quarter-to-quarter) for a firm. For example, Apple's quarterly conference calls are always held from 5 p.m. to 6 p.m. Eastern Standard Time.⁵

Since management ability uncertainty creates demand for information by outsiders, we argue that forward-looking disclosures are made in response to this demand. Unlike disclosures about known past events, forward-looking statements help resolve uncertainty about future trends, plans, and actions that likely have a material impact on a company (SEC 1989, 2003). We use Muslu, Radhakrishnan, Subramanyam, and Lim's (2015) methodology to identify forward-looking sentences in earnings conference calls. The number of words in a CEO's forward-looking statements scaled by the total number of words spoken by the CEO is our measure of each CEO's forward-looking disclosures. We similarly calculate all other non-CEO executives' forward-looking disclosures using their statements in earnings calls.

Using a longitudinal research design and nonlinear transformations of CEO tenure, we examine how CEOs' forward-looking disclosures change over their tenure.⁶ In multivariate tests that include firm characteristics (e.g., performance, firm size, growth potential), changes in operating environment (e.g., volatility, capital expenditures), and firm-CEO and year-quarter fixed effects, we find a significant decline in CEOs' forward-looking disclosures over their tenure.⁷ This effect is economically

² For example, Marissa Mayer, CEO of Yahoo! appointed in July 2012, in her first earnings conference call, discussed "the vision and direction for Yahoo! moving forward . . . Our goals are simple: execute faster, return value to our shareholders, attract the best talent and make Yahoo! the absolute best place to work" (YHOO 2012).

³ Our setting can also be used to examine temporal changes in disclosure styles of other firm officers (e.g., CFO, COO). However, empirical power is limited due to significantly shorter tenures of non-CEO officers, difficulty in precisely identifying their appointment dates, and their limited or infrequent participation in quarterly conference calls.

⁴ For instance, mandatory annual and quarterly Securities and Exchange Commission (SEC) reports are often criticized for being repetitive and polished by lawyers (see Monga and Chasan 2015). While lawyers are likely involved in earnings calls, a large portion of such disclosures are both strategic (i.e., to preempt questions) and spontaneous (i.e., driven by analysts' questions).

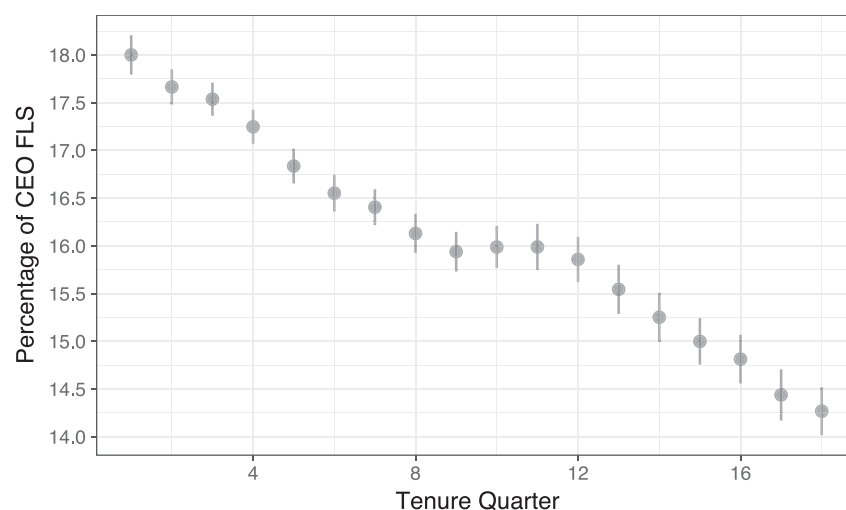
⁵ Analyzing a large sample of earnings call times and duration available on <https://earningscast.com/> (12,956 observations for 2,397 unique companies), we find that the median difference in duration between consecutive conference calls for a company is around six minutes. In terms of spoken words, we find that the autocorrelation of CEO spoken words is 0.75 and of non-CEO spoken words is 0.78. This suggests that executive speech in earnings conference calls is largely time-invariant. However, although there is little variation in call duration within a firm, there is substantial variation across firms. Accordingly, we include firm-CEO fixed effects in our empirical specifications.

⁶ Appendix A provides an example of two earnings call transcripts, one at the beginning of the Dominion Resources CEO's tenure and the other at six years into his tenure. We highlight future- and past-oriented sentences in italic and bold, respectively. The example illustrates that earnings calls are primarily used as a medium to discuss future plans and actions and to provide context for the current performance.

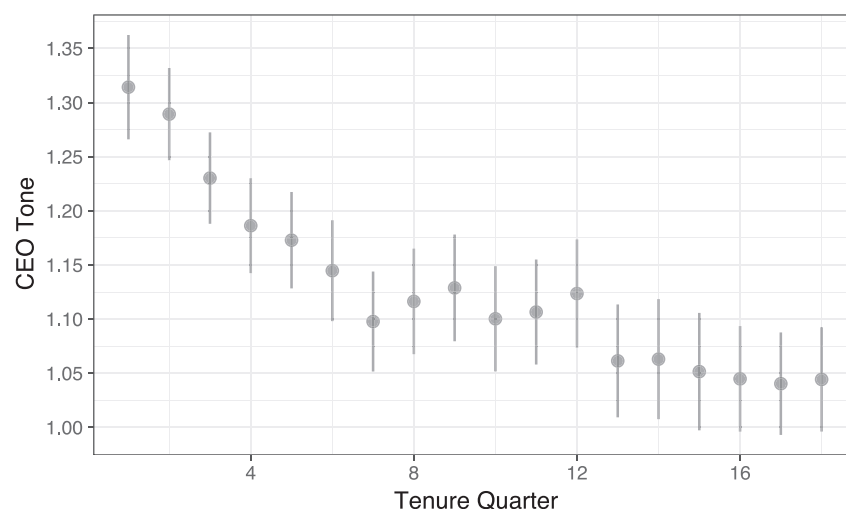
⁷ As noted by Pan et al. (2015), CEO turnovers typically lead to significant changes in firm operations (e.g., restructurings) that exacerbate uncertainty. Accordingly, our empirical analyses include measures of firm performance, changes in operations, volatility, and other relevant factors.

FIGURE 1
CEO Tenure and CEO Disclosure Style

Panel A: CEO FLS



Panel B: CEO Tone



This figure plots point averages of fitted values of the percentage of CEO-specific forward-looking statements (*CEOFLS*, Panel A) and CEO-specific tone (*CEOTone*, Panel B) in earnings conference calls for the first 18 tenure-quarters. Fitted values of *CEOFLS* and *CEOTone* are estimated using spline regressions that control for firm performance, annual stock return, unexpected earnings, size, market-to-book, leverage, earnings and return volatility, special items, discontinued operations, capital expenditures, R&D expense, advertising expense, analyst following, and firm-CEO and year-quarter fixed effects.

All variables are defined in Appendix B.

meaningful: relative to the first tenure-quarter, the proportion of forward-looking information disclosed by a CEO decreases, on average, by 7–7.6 percent after the first year and 13.43–22.6 percent over the first four years of their tenure, based on the empirical specification.⁸ Figure 1, Panel A graphically displays the quarter-over-quarter decline in CEOs' forward-looking disclosures over their tenure. Further, we find no significant relation between CEO tenure and forward-looking disclosures

⁸ The economic magnitude of results is determined relative to the average CEO forward-looking disclosures in the first tenure-quarter (equal to 17.957 percent in our sample).

made by the other participating non-CEO executives over their CEO's tenure. Thus, the decline in CEOs' forward-looking disclosures over their tenure is only CEO-specific.

The decline in the amount of CEOs' forward-looking disclosures over their tenure is consistent with the Bayesian learning model about *CEO ability*, because the uncertainty about CEO leadership is the highest at the beginning of a CEO's appointment and declines with CEO tenure (see Pan et al. 2015). This uncertainty is (1) due to the unobservability of actions and private knowledge, (2) because the "match" between a firm and its CEO is unknown to both the CEO and outside stakeholders and can only be ascertained over time, and (3) due to the lack of CEO history at the firm. As outsiders' beliefs about firm management become increasingly precise over time, the value of newly disclosed information declines, thereby lowering their demand for information.^{9,10}

In a related study, Pan et al. (2015) use a Bayesian learning model to examine the relation between CEO tenure and investors' perceptions of managerial ability uncertainty, as reflected in a firm's stock return volatility. They find that stock return volatility is greatest at the beginning of CEO tenure and gradually declines over time. They interpret this result as evidence of greater uncertainty about managerial ability around the CEO appointment, and the decline in volatility over CEO tenure as evidence of investors learning about managerial ability over time. While Pan et al. (2015) focus on the way this learning affects firms' return volatility, we study how CEOs' voluntary disclosures are affected by the market's uncertainty about CEO ability. Consequently, our work studies how managerial ability uncertainty, as proxied by CEO tenure, relates to CEOs' disclosures, rather than whether disclosures resolve managerial ability uncertainty.

To provide further evidence that the decline in CEOs' forward-looking disclosures over tenure is driven by the reduction in ability uncertainty rather than firm-specific factors, we examine the relation between disclosure style and managerial characteristics that potentially affect managerial ability uncertainty. Consistent with Pan et al. (2015), we posit that there is greater uncertainty regarding the ability of externally hired CEOs than internally promoted CEOs because the former lack firm-specific experience. Similarly, uncertainty about a CEO as a firm leader is likely higher for those without prior executive experience. In addition, there is potentially greater uncertainty regarding the ability of younger CEOs because of their shorter career histories than for older and more experienced managers. Consistent with our predictions that high uncertainty about CEO ability increases the demand for and supply of information, we find that externally hired CEOs and those without prior executive experience (CEO, CFO, or COO) provide more forward-looking information than their internally hired or experienced counterparts. Further, for these CEOs, we find a steeper decline in forward-looking disclosures with tenure, suggesting that CEOs with greater ability uncertainty provide more information to the market early in their tenure. However, we find no differences between forward-looking disclosures made by younger and older CEOs.

Managerial ability uncertainty also creates career concerns for CEOs, since it affects the precision of outsiders' (capital and labor markets) beliefs about their ability and the extent to which beliefs are updated in light of new information. While outsiders use various signals to learn about managerial ability, periodic voluntary disclosures, such as earnings announcements and accompanying conference calls, are important information channels. We argue that CEO tone (optimistic versus pessimistic), a metric that has been shown to predict future firm performance (see Li [2010b] and Loughran and McDonald [2016] for reviews), is a relevant signal of CEO ability to the market. To examine changes in CEOs' disclosure optimism, we examine how CEOs' tone changes over their tenure. The number of positive minus the number of negative words, scaled by the total words spoken by a CEO, is our measure of the CEO's relative optimism. We similarly calculate all other non-CEO executives' optimism using their statements in earnings calls.

Unlike forward-looking statements that measure *what* information CEOs disclose, the tone of CEOs' statements measures *how* they present information. Using the same longitudinal research design as before, we find a significant decline in CEOs' optimism in conference calls over their tenure. Specifically, we document that relative to the first tenure-quarter, there is a 13.61 percent–16.01 percent decrease in CEOs' linguistic tone after the first year of tenure, and a 23.97 percent–38.35 percent decrease over the first four years of their tenure, based on the empirical specification employed.¹¹ Figure 1, Panel B shows the quarter-over-quarter decline in CEOs' tone in earnings calls over their tenure. We do not observe a similar decline in non-CEO executives' optimism over their CEO's tenure, suggesting that the decline is only CEO-specific. These results suggest that CEOs' disclosures exhibit greater optimism early in their tenure, but this effect diminishes over the course of their employment.

The decline in CEOs' disclosure optimism over their tenure is consistent with the *attenuation* of CEOs' *career concerns* over time. Since labor market participants cannot directly observe CEOs' ability to enhance firm value, they revise their beliefs

⁹ See Hermalin and Weisbach (1998) for an illustration of how beliefs increase in precision over time as new information is revealed.

¹⁰ In contrast, our results do not support the hypothesis that CEOs' information accumulation over time affects their disclosures. The increasing temporal accumulation of information by CEOs would lead firm owners to increasingly demand that information be revealed (i.e., resulting in more disclosure over tenure) or, alternatively, increasingly penalize managers for withholding information (Verrecchia 1983).

¹¹ The reported economic magnitudes are calculated relative to the average value of CEO linguistic tone measure in the first tenure-quarter (equal to 1.168 in our sample).

in light of information contained in periodic reports, which affects a CEO's future wage, autonomy, and dismissal likelihood (Gibbons and Murphy 1992a; Hermalin and Weisbach 1998).¹² Consequently, CEOs have an incentive to favorably influence the market's assessment of their ability by presenting their performance and strategies in the best possible light (see the signal-jamming framework in Stein [1989] and Holmstrom [1999]). The incentive to favorably influence the market's perception of their ability is greatest early in their tenure, and diminishes over time as the market's beliefs increase in precision due to the accumulation of performance information (Holmstrom 1999).¹³ To favorably affect the market's beliefs about their ability, managers work harder earlier in their careers (Holmstrom 1999), overstate reported earnings (Ali and Zhang 2015), and manage their earnings guidance (Pae, Song, and Yi 2016; Baginski, Campbell, Hinson, and Koo 2018; Shaikh 2017). Our results contribute to this literature as we document that CEOs, early in their tenure, use more positive (or less negative) language to favorably influence outsiders' perceptions of their ability. Over the course of their appointments, as the precision of outsiders' beliefs about a CEO's ability increases, managerial career concerns ebb, diminishing these incentives to influence outsiders' perceptions through their disclosures. We also find that market reactions to CEO tone are lower in early tenure years. This is consistent with the conjecture that in equilibrium, outsiders are not fooled by CEOs' signal-jamming (Stein 1989; Holmstrom 1999): they correctly anticipate and rationally discount the excessive optimism at the beginning of a CEO's appointment.

To provide further evidence that the decline in CEOs' optimism over their tenure is driven by their career concerns, we examine the relation between CEO tone and CEO characteristics that potentially affect their career concerns. We posit that career concerns are higher for externally hired CEOs, CEOs without prior executive experience, and younger CEOs, leading to greater optimism in their disclosures earlier in their tenure (Gibbons and Murphy 1992b; Pan et al. 2015). We find that younger CEOs exhibit greater optimism in their disclosures than their older counterparts. However, we do not observe a statistically significant association between CEOs' optimism, their hiring type (external versus internal), and their prior executive experience.

We conduct a series of additional analyses and robustness tests to ensure that our results are unaffected by our empirical design choices. First, we partition CEO tone into its positive and negative components (i.e., usage of positive versus negative words) and find that the positivity in CEOs' language declines over tenure, whereas negativity increases in tenure. We also examine the relative CEO tone in forward-looking and non-forward-looking statements, and find that the decline in tonal positivity is largely in the non-forward-looking statements that correspond to contemporaneous and past performance disclosures.

Second, we estimate our results on subsamples of new CEOs that were appointed following performance-related and non-performance-related turnovers. We find that while CEOs' forward-looking disclosures and optimism are negatively associated with their tenure in both subsamples, these relations are more pronounced among CEOs that are appointed following a performance-related turnover. These results attest to the greater uncertainty due to changes in firm strategy and operations, which diminishes over time, about a new CEO's ability following a forced CEO turnover affecting an incoming CEO's disclosure style.

We also reexamine the dynamics of CEO disclosure style using a reduced sample of "surviving" CEOs, i.e., those CEOs who begin their appointments after 2005 and hold office for at least 18 quarters. This allows us to examine disclosure dynamics for the same (constant) set of CEOs over time. We find that the negative relation between CEOs' tenure and their disclosure style maintains for these CEOs. This analysis shows that our results are not driven by either CEOs who are not in office long enough to estimate any meaningful temporal trends or by the lower "survival" likelihood of low-ability CEOs. Next, for a smaller sample of CEOs for whom we can collect their predecessors' conference call transcripts two years prior to their appointment, we compare their disclosure styles to those of their predecessors. We identify a sawtooth pattern for CEO disclosure styles: forward-looking disclosures and tone are significantly lower in the years before a CEO appointment, spike upward in the first year of the appointment, and then resume the downward trend documented earlier.

Finally, we generalize our findings by examining CEOs' disclosures in the presentation and question-and-answer (Q&A) sections of the earnings call, and using a cross-sectional research design on a sample of all firm-CEO observations with available earnings call transcripts from 2006 to 2014. As before, we find that CEO-specific forward-looking disclosures and CEO-specific optimism are negatively related to CEO tenure. In summary, finding consistent results across multiple predictions and specifications makes it unlikely that our results are attributable to alternative explanations or specific design choices.

¹² Nagar (1999) argues that voluntary disclosures impose risk for a disclosing manager since the labor market's inference is uncertain.

¹³ The secular decline in CEO relative optimism is inconsistent with models that potentially characterize tone as cheap-talk (Crawford and Sobel 1982). Cheap-talk occurs only when the objectives of the manager (i.e., sender) are not "too different" from the market (i.e., receiver). Since differences in objectives should decline over CEO tenure, cheap-talk models imply that we are more likely to observe relative optimism in CEO tone later in their tenure when career concerns are lower.

Our paper documents the dynamics of CEOs' disclosure styles by demonstrating the temporal decline in both the forward-looking information and optimism of CEO-specific disclosures over their tenure. We interpret the temporal decline in CEOs' forward-looking disclosures over their tenure as consistent with investors learning about uncertain managerial ability leading to a decline in their demand for information. Nevertheless, this evidence can also be interpreted as consistent with CEOs speaking about the future at the start of their appointments because they lack past performance to address. Our evidence on the decline in CEO optimism is consistent with ameliorating managerial career concerns over time as investors' beliefs about managerial ability become increasingly precise.

Our work supplements a large body of literature on disclosure by showing that, in addition to firm-specific information, disclosures provide insight about firm managers and their incentives. There is a long tradition in economic science of empirically examining how CEOs affect their firms' business operations, economic performance, financial reporting, information environment, and firm valuation (Johnson, Magee, Nagarajan, and Newman 1985; Weisbach 1995; Clayton, Hartzell, and Rosenberg 2005; Bushman, Dai, and Wang 2010; Ali and Zhang 2015). Inspired by theoretical models that incorporate Bayesian learning about managerial ability (Holmstrom 1999; Gibbons and Murphy 1992a; Hermalin and Weisbach 1998), recent empirical work has examined the dynamic consequences to a firm's information environment affected by management changes and documented a negative relation between CEO tenure and firm-level uncertainty (Taylor 2013; Pan et al. 2015).¹⁴ We complement this literature by identifying how uncertainty about their managerial ability affects the manner in which CEOs disclose relevant information to market participants. Consistent with the implications of prior Bayesian learning and managerial career concern models, our empirical results show that CEOs' disclosures are more future-oriented and optimistic early in their appointments, when managerial uncertainty and career concerns are heightened, but gradually decline in their tenure as managerial ability uncertainty attenuates.

II. DATA AND METHODOLOGY

To empirically examine the dynamics of CEO disclosure style, we construct a sample of CEO-specific disclosures in quarterly earnings conference calls that accompany earnings announcements. In contrast to written formal disclosures mandated by the SEC (e.g., 10-K and 10-Q), earnings conference calls are public spoken events intended to remotely connect a company's top management with participating analysts and investors.¹⁵ Since conference calls are periodic events that contain managers' prepared statements and spontaneous responses to participants' questions, they provide an ideal setting for examining CEOs' disclosure styles over their tenure.

We obtain quarterly earnings conference call transcripts from Seeking Alpha (at: <https://seekingalpha.com/>) for the period from 2006 to 2014.¹⁶ Seeking Alpha is one of the largest investor-oriented websites in the United States that covers a broad range of publicly traded companies and provides free access to earnings conference call transcripts. We use XML tags in the call transcripts to identify call participants and portions of text corresponding to their speech. This allows us to identify the text corresponding to each CEO and non-CEO company officer in both the presentation and questions-and-answers sections of the call.

We collect data on CEOs and their tenure from Execucomp. Execucomp provides information on the top five executives of current Standard & Poor's (S&P) 1500 firms or firms that were part of the S&P 1500 in prior years. The database provides information on the start and termination dates for each CEO, which enables us to measure CEO tenure on a quarterly basis and match with quarterly earnings conference calls. We use the employment start date (*BECAMECEO*) and quarter-end date (*DATADATE*) to calculate CEO tenure, as the quarterly difference between the two dates ($DATADATE - BECAMECEO$) if CEO termination date (*LEFTOFC*) is either missing or occurs after the earnings announcement date. Dikolli, Mayew, and Nanda (2014) argue that measuring CEO tenure on a quarterly basis enables more precise estimates of tenure-related trends, given that the median U.S. public company CEO tenure is less than five years. We exclude "interim" CEOs by removing observations that correspond to CEOs that have held office for less than two quarters. This additional constraint limits our sample to 41,776 observations for the years 2006–2014.

We obtain financial statements, analyst forecasts, and market data from Compustat, I/B/E/S, and CRSP, respectively. For each quarterly observation, we calculate firm earnings surprise, loss, return on assets, annual stock return, market capitalization, market-to-book ratio, leverage, earnings volatility, return volatility, magnitude of special items and discontinued operations, change in capital expenditures, change in research and development (R&D) expense, change in advertising expense, and

¹⁴ Taylor (2013) examines how learning about CEO ability affects the bargaining power between firms and CEOs, while Pan et al. (2015) focus on the way in which learning about CEO ability affects firms' return volatility. Our analysis builds on the same theories as in Taylor (2013) and Pan et al. (2015), and pins down the way in which CEO ability uncertainty affects disclosures.

¹⁵ Typically, a conference call starts with a brief introduction of the management team present on the call and a legal disclaimer about forward-looking statements. Then company executives (CEO, CFO, etc.) give an overview of the operating performance for the quarter just ended and provide information on future plans and operations. After the introductory statements by managers, the call is opened for questions from analysts and investors.

¹⁶ Seeking Alpha was founded in 2004, but a comprehensive coverage of firms on its website started in 2006–2007.

analyst following. To estimate a firm's quarterly earnings surprise, we use the most recent analyst consensus forecast of one- or two-quarters-ahead earnings issued or reviewed in the last 60 days before each earnings announcement. Merging the financial data with our CEO tenure data yields a sample of 32,081 unique CEO-quarter observations. To complete our dataset, we merge the CEO and firm financial data with the earnings conference call transcript data. Each transcript contains identifying information about a company, including its name, ticker, and the date of the earnings call, which we match to company names, tickers, and earnings announcement dates in the Compustat quarterly file.¹⁷ In our sample, most companies host conference calls on the day of their earnings announcement (around 80 percent) or on the following day (around 18 percent), and a few companies host the call within one week following the earnings announcement (around 2 percent). We only retain observations where a firm's CEO participates in the call. As a result, our sample is reduced to 18,660 CEO tenure-quarter observations for 1,876 unique CEOs. Further, our analyses focus on examining disclosure styles of newly hired CEOs that begin their appointments as CEO of publicly listed U.S. firms after 2005 and for whom we can obtain quarterly earnings conference call transcripts from the beginning of their appointment.¹⁸ We also require at least two observations per CEO. This results in a sample of 5,581 observations of 573 unique CEOs appointed after 2005. Panel A of Table 1 summarizes our sample selection.

In Panel B of Table 1, we tabulate the number of all earnings call transcripts and the transcripts where a CEO participates in the call for the first 18 CEO tenure-quarters. We observe that CEO participation in earnings conference calls does not change with their tenure, as the proportion of calls with CEO participation remains roughly the same throughout CEO tenure (around 87 percent).¹⁹ This provides confidence that excluding observations with no CEO participation in earnings calls does not bias our inferences. Finally, in Panel C of Table 1, we tabulate new CEO appointments and all observations with available earnings call transcripts for each calendar year in our sample. The temporal variation in the number of observations is largely driven by Seeking Alpha's coverage and our requirements of the availability of post-hiring conference call transcripts.

Our main dependent variables of interest are the measures of CEO-specific forward-looking disclosures and CEO-specific linguistic tone in earnings conference calls. While forward-looking disclosures measure CEOs' supply of information about firm prospects to outside stakeholders, linguistic tone captures CEOs' relative optimism about their firms' performance and prospects. We follow the methodology in Muslu et al. (2015) to extract forward-looking statements (FLS) in earnings calls. Specifically, we use a broad list of future-oriented keywords and phrases to distinguish forward-looking statements from statements referring to concurrent or past events (see Muslu et al. 2015, Appendix).²⁰ Accordingly, we identify forward-looking sentences as those that contain any of the following phrases or references:

1. Keywords that are commonly used to refer to the future: "will," "future," "next fiscal/month/period/quarter/year," "incoming fiscal/month/period/quarter/year," "coming fiscal/month/period/quarter/year," "upcoming fiscal/month/period/quarter/year," "subsequent fiscal/month/period/quarter/year," "following fiscal/month/period/quarter/year."

Example: "Our infrastructure growth program continues to provide the foundation for sustained earnings growth in the future" (Dominion Resources Q3 2010 Earnings Call).

2. Verb conjugations that relate to the future: "aim," "anticipate," "assume," "commit," "estimate," "expect," "forecast," "foresee," "hope," "intend," "plan," "project," "seek," "target."

Example: "We plan to file for approval of this plant with the Virginia State Corporation Commission" (Dominion Resources Q3 2010 Earnings Call).

3. References to future periods relative to the period of earnings call.

Example: "The plant is scheduled for commercial operation in 2012" (Dominion Resources Q3 2010 Earnings Call).

We calculate the CEO-specific forward-looking disclosures measure as the number of words in CEO forward-looking statements divided by the total number of CEO words:²¹

¹⁷ To ensure the accuracy of the matching, we perform extensive manual checks of matched company names and tickers.

¹⁸ We retain CEO-quarter observations for CEOs that have at least one observation in Execucomp in their first year to mitigate the left-censoring problem. This restriction is necessary for a longitudinal analysis of CEO disclosure style in event-time following CEO appointment. In our cross-sectional analysis, we relax these restrictions and conduct our tests on a larger sample of all CEO-quarterly observations (see Section IV). All our inferences remain the same.

¹⁹ We confirm this univariate result using a regression analysis (untabulated), where CEO participation is regressed on CEO tenure and firm characteristics.

²⁰ Muslu et al. (2015) use a text-parsing software to extract forward-looking sentences from the management discussion and analysis (MD&A) section in annual reports and validate their approach by comparing their results against forward-looking statements identified by human annotators in a set of randomly selected MD&A documents.

²¹ We also calculate an alternative measure of CEO-specific forward-looking disclosures, *CEOFLSAIt*, as the number of words in CEO forward-looking statements divided by the number of words in all company executives' statements. All our inferences remain the same.

TABLE 1
Sample Description

Panel A: Sample Selection

CEO tenure-quarter observations from Execucomp (years 2006–2014) of CEOs who have held their CEO positions for more than two quarters.	41,776
CEO tenure-quarter observations with required Compustat and CRSP data.	36,393
CEO tenure-quarter observations with required I/B/E/S data.	32,081
CEO tenure-quarter observations with conference call transcripts where a CEO participates in the conference call.	18,660
CEO tenure-quarter observations with newly hired CEOs after 2005 for whom there is at least one observation in their first tenure year in the merged sample (with the minimum of two observations per CEO).	5,581

This panel summarizes our sample selection. The sample covers the period of 2006–2014.

Panel B: CEO Participation in Earnings Conference Calls: First 18 Tenure-Quarters

CEO Tenure Quarters	Number of Transcripts	Number of Transcripts with CEO Participation	Proportion of Transcripts with CEO Participation
1	324	288	89%
2	416	366	88%
3	479	402	84%
4	455	400	88%
5	448	385	86%
6	384	330	86%
7	373	328	88%
8	342	301	88%
9	325	283	87%
10	275	242	88%
11	257	226	88%
12	246	209	85%
13	203	179	88%
14	186	162	87%
15	187	163	87%
16	172	153	89%
17	164	146	89%
18	161	140	87%
Total	5,397	4,703	

This panel shows the summary statistics of all earnings conference call transcripts available on <https://seekingalpha.com/>, and proportion of transcripts where CEOs participate in earnings calls for the first 18 CEO tenure-quarters. Sample is conditional on the availability of all Compustat, CRSP, Execucomp, I/B/E/S, and BoardEx variables required for the analysis.

(continued on next page)

$$CEOFLS = 100 \times \frac{CEO \text{ Forward-Looking Words}}{CEO \text{ Total Words}}.$$

Similarly, we calculate the forward-looking disclosures measure for all other non-CEO officers present on the call, *NonCEOFLS*, as the number of words in their forward-looking statements divided by the number of words in all their statements.

To measure CEO-specific linguistic tone in earnings calls, we use all CEO statements (i.e., both forward-looking and other), since our intent is to analyze CEOs' relative optimism regarding their firms' current and future operations and performance. However, in additional tests, we distinguish between CEO tone corresponding to forward-looking statements and to statements discussing past or concurrent performance. We use Loughran and McDonald's (2011) financial sentiment dictionary (hereafter, L&M dictionary) to identify positive and negative words in CEO-specific disclosures in earnings calls. The L&M dictionary was developed to analyze qualitative corporate disclosures and is now widely used among researchers to gauge linguistic tone (see, for example, Feldman, Govindaraj, Livnat, and Segal 2010; Rogers et al. 2011; Huang, Teoh, and

TABLE 1 (continued)

Panel C: New CEO Appointments and CEO Tenure-Quarter Observations by Year

Year	New CEO Appointments with Available Transcripts	All CEO Tenure-Quarters with Available Transcripts
2006	22	45
2007	79	219
2008	98	557
2009	63	629
2010	56	620
2011	66	783
2012	100	1,143
2013	89	1,287
2014	2	298
Total	575	5,581

This panel shows the summary statistics of newly hired CEOs each year with available earnings conference call transcripts on <https://seekingalpha.com> (second column). The third column summarizes all CEO tenure-quarter observations with available earnings call transcripts. Sample is conditional on the availability of all Compustat, CRSP, Execucomp, I/B/E/S, and BoardEx variables required for the analysis.

Zhang 2014; Chen, De, Hu, and Hwang 2014). We calculate CEO linguistic tone in earnings conference calls as the number of positive words minus the number of negative words divided by the total number of words spoken by a CEO:

$$CEOTone = 100 \times \frac{CEO \text{ Positive Words} - CEO \text{ Negative Words}}{CEO \text{ Total Words}}.$$

Similarly, using statements made by non-CEO officers, we calculate the linguistic tone of other (non-CEO) company executives (*NonCEOTone*). Intuitively, *CEOTone* (*NonCEOTone*) measures the relative optimism expressed by a CEO (non-CEO firm officers) when communicating with investors and analysts in conference calls.

To illustrate our method, we display *CEOFLS* and *CEOTone* measures for Thomas Farrell, CEO of Dominion Resources, appointed in January 2006, for the first five years of his tenure. In his first year as CEO, Thomas Farrell provides substantial forward-looking disclosures, as indicated by the 35.9 percent value for *CEOFLS*, as well as relatively high disclosure optimism, as indicated by the 1.41 value for his linguistic tone. In subsequent years, both the forward-looking information and the disclosure positivity decline over his tenure with the firm. This example provides an illustration of our empirical measures and anecdotal evidence of changes in CEOs' disclosure styles over their tenure.

Call Period	CEOFLS	CEOTone
2006 Q3 (1st tenure year)	35.99% (1,182 FL words)	1.41 (47 positive and 8 negative words)
2007 Q3 (2nd tenure year)	28.96% (875 FL words)	0.35 (35 positive and 22 negative words)
2008 Q3 (3rd tenure year)	24.07% (762 FL words)	0.49 (43 positive and 30 negative words)
2009 Q3 (4th tenure year)	17.43% (402 FL words)	0.71 (20 positive and 10 negative words)
2010 Q3 (5th tenure year)	11.98% (168 FL words)	0.06 (19 positive and 18 negative words)

Table 2 presents descriptive statistics for our two measures of CEO disclosure style (forward-looking information and tone), CEO tenure and age, and other control variables for 5,581 quarterly earnings conference calls over the period 2006 to 2014. The mean (median) of *CEOFLS* is 16.08 percent (15.25 percent), while the mean (median) of *NonCEOFLS* is 16.37 percent (15.59 percent). This indicates that a large portion of the earnings conference call is forward-looking. The mean (median) of *CEOTone* is 1.14 (1.12), while the mean (median) of *NonCEOTone* is 0.38 (0.39). This suggests that CEOs are, on average, more optimistic in their statements than other participating non-CEO officers. Also, consistent with Davis et al. (2015), we find that CEOs and other executives tend to use more positive than negative words in earnings conference calls. The mean (median) CEO age in our sample is 54 (54) years. Finally, the mean (median) CEO current tenure (i.e., the number of quarters since the appointment) and CEO maximum tenure (i.e., the number of all tenure-quarters) are ten (eight) and 18 (18) quarters, respectively.

Table 3 reports Pearson correlations for our variables of interest. We find a significant negative correlation of 14 percent ($p < 0.01$) between CEO tenure and the proportion of forward-looking statements made by a CEO. In contrast, CEO linguistic tone is unconditionally positively correlated with tenure (correlation of 4 percent, $p < 0.01$). However, linguistic tone is also

TABLE 2
Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Q1	Q3
<i>CEOFLS</i>	16.080	15.250	6.478	11.401	20.105
<i>CEOFLSAlt</i>	8.436	7.837	4.281	5.322	10.868
<i>CEOTone</i>	1.140	1.120	0.789	0.631	1.662
<i>CEOWords</i>	8.044	8.109	0.479	7.766	8.392
<i>ExecWords</i>	8.728	8.782	0.320	8.559	8.937
<i>NonCEOFLS</i>	16.376	15.593	6.703	11.816	20.244
<i>NonCEOTone</i>	0.384	0.393	0.722	−0.055	0.840
<i>CEOTenure</i> (in quarters)	10.238	8.000	7.221	4.000	15.000
<i>CEOAge</i> (in years)	54.002	54.000	5.906	50.000	58.000
<i>CEOTenure</i> (logged)	2.022	2.079	0.856	1.386	2.708
<i>CEOAge</i> (logged)	3.983	3.989	0.112	3.912	4.060
<i>UE</i>	0.001	0.001	0.008	−0.000	0.002
<i>Loss</i>	0.150	0.000	0.357	0.000	0.000
<i>ROA</i>	0.011	0.011	0.026	0.003	0.021
<i>Return</i>	0.150	0.123	0.420	−0.099	0.344
<i>Size</i>	8.479	8.498	1.519	7.475	9.469
<i>MTB</i>	1.676	1.382	0.902	1.065	1.977
<i>Leverage</i>	0.191	0.172	0.150	0.062	0.289
<i>RetVol</i>	0.025	0.022	0.013	0.015	0.032
<i>EarnVol</i>	0.020	0.010	0.026	0.005	0.024
<i>SpecialItems</i>	−0.175	−0.009	0.325	−0.212	0.000
<i>DiscOperations</i>	0.000	0.000	0.025	0.000	0.000
Δ CAPX	−0.017	0.000	2.464	−0.426	0.544
Δ R&D	0.025	0.000	1.719	0.000	0.000
Δ ADV	−0.028	0.000	1.758	0.000	0.000
<i>Following</i>	2.128	2.303	0.866	1.609	2.773
Observations	5,581				

This table shows the descriptive statistics for variables used in our main analyses for the period 2006–2014. All variables are defined in Appendix B.

significantly positively correlated with firm performance (ROA, stock returns, and unexpected earnings). Consequently, the univariate correlations are potentially misleading regarding the relation between CEO tenure and disclosure tone since firm performance is a confounding variable.

To further illustrate the variation in CEOs' and non-CEO officers' forward-looking disclosures, linguistic tone, and firm performance over their tenure, in Table 4, we provide univariate means for these measures for their first 18 tenure-quarters.²² The tabulated results show a decline in CEOs' forward-looking statements over their tenure, but we do not observe a corresponding decline for non-CEO officers. We also report univariate means for the length of executive disclosures, measured as the natural logarithm of CEOs' and all executives' total spoken words in the conference call. We do not observe any significant increasing or decreasing patterns in either of these measures. This suggests that the negative correlation between *CEOFLS* and *CEOTenure* is not due to changes in *CEOWords*, which we use as a denominator to calculate *CEOFLS*. Importantly, we observe a similar relation with *CEOTenure* for an alternative measure of *CEOFLS*, *CEOFLSAlt* (i.e., when all executives' words is the denominator).

Further, the univariate means are not suggestive of a relation between CEO and non-CEO linguistic tone and CEO tenure, but we do observe a positive increasing trend in firm performance and CEO tenure as under-performing managers exit the sample over time.²³ Because firm performance potentially confounds the relation between CEO tone and tenure, we

²² The total number of observations in Table 4 is less than our main sample since we have an additional (not included in the table) 1,018 CEO tenure-quarter observations in tenure-quarters exceeding 18.

²³ In addition, higher firm performance in later quarters of CEO tenure is likely attributable to the post 2007–2009 financial crisis period. Therefore, to estimate meaningful trends in CEOs' disclosure styles over their tenure, it is important to include performance measures and time effects in a regression setting.

TABLE 3
Correlation Table

Panel A: Correlation Variables *CEOFLS* to *Size*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) <i>CEOFLS</i>	1.00								
(2) <i>CEOFLSAlt</i>	0.70***	1.00							
(3) <i>CEOTone</i>	-0.06***	-0.10***	1.00						
(4) <i>CEOTenure</i>	-0.14***	-0.14***	0.04***	1.00					
(5) <i>UE</i>	-0.03**	-0.01	0.11***	0.04***	1.00				
(6) <i>Loss</i>	0.10***	0.06***	-0.08***	-0.09***	-0.25***	1.00			
(7) <i>ROA</i>	-0.10***	-0.04***	0.13***	0.07***	0.22***	-0.58***	1.00		
(8) <i>Return</i>	-0.08***	-0.05***	0.18***	0.13***	0.08***	-0.19***	0.22***	1.00	
(9) <i>Size</i>	-0.11***	-0.20***	0.07***	0.16***	0.01	-0.29***	0.24***	0.05***	1.00
(10) <i>MTB</i>	-0.07***	-0.02	0.18***	0.01	0.01	-0.15***	0.38***	0.18***	0.21***
(11) <i>Leverage</i>	0.06***	0.03**	-0.07***	0.04***	-0.04***	0.07***	-0.09***	0.00	-0.01
(12) <i>EarnVol</i>	0.04***	0.10***	0.06***	-0.05***	0.01	0.24***	-0.17***	0.01	-0.31***
(13) <i>RetVol</i>	0.12***	0.12***	-0.12***	-0.19***	-0.07***	0.38***	-0.24***	-0.24***	-0.48***
(14) <i>SpecialItems</i>	-0.03**	-0.04***	-0.04***	0.07***	-0.02*	-0.14***	0.12***	0.06***	0.10***
(15) <i>DiscOperations</i>	-0.00	-0.01	-0.00	-0.02*	-0.00	-0.01	-0.00	0.04***	0.08***
(16) Δ CAPX	0.01	-0.01	-0.03**	-0.02	-0.02	-0.00	-0.00	-0.10***	0.02
(17) Δ R&D	-0.00	-0.02	0.01	-0.01	-0.04***	0.05***	-0.04***	-0.04***	-0.02*
(18) Δ ADV	0.02	0.01	-0.00	-0.02	-0.00	-0.01	-0.01	-0.01	0.01
(19) <i>Following</i>	-0.10***	-0.16***	0.01	0.12***	0.01	-0.11***	0.11***	-0.05***	0.55***

Panel B: Correlation Variables *MTB* to *Following*

	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
(10) <i>MTB</i>	1.00								
(11) <i>Leverage</i>	-0.17***	1.00							
(12) <i>EarnVol</i>	0.03**	-0.04***	1.00						
(13) <i>RetVol</i>	-0.03**	-0.03**	0.33***	1.00					
(14) <i>SpecialItems</i>	0.05***	-0.06***	-0.19***	-0.15***	1.00				
(15) <i>DiscOperations</i>	0.01	-0.01	-0.02*	-0.05***	0.03**	1.00			
(16) Δ CAPX	-0.01	-0.02*	0.02*	-0.03**	0.01	0.01	1.00		
(17) Δ R&D	0.02	-0.01	0.04***	0.04***	-0.04***	0.01	0.06***	1.00	
(18) Δ ADV	-0.08***	-0.01	0.00	-0.01	-0.03**	0.01	-0.01	0.02	1.00
(19) <i>Following</i>	0.05***	-0.00	-0.10***	-0.11***	0.06***	0.05***	0.00	-0.02	0.03*

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows Pearson correlations (with significance levels) between our main variables of interest. All variables are defined in Appendix B.

conduct conditional tests that include covariates related to firm performance, time effects, and other firm and managerial characteristics. To account for the potential survivor bias induced by the exit of under-performing CEOs, we also conduct our analysis on a smaller subsample consisting of only those CEOs that are appointed after 2005 and survive at least 18 quarters (see Section IV).

III. EMPIRICAL TESTS AND RESULTS

In this section, we examine the relation between CEOs' disclosure styles and their tenure in a panel regression setting. In the first subsection, we examine the relation between the CEO-specific forward-looking disclosures in earnings calls and CEO tenure. In the second subsection, we analyze the relation between CEO-specific linguistic tone in earnings calls and CEO tenure.

TABLE 4
Disclosure Style in Earnings Conference Calls and Firm Performance over CEO Tenure, Univariate Evidence

Tenure Qtr.	Obs.	FLS			Tone		Call Length		Performance		
		CEO- FLS	CEO- FLSAlt	NonCEO- FLS	CEO- Tone	NonCEO- Tone	CE- Words	Exec- Words	ROA	UE	Return
1	288	17.9570	9.6006	16.3587	1.1682	0.2715	8.0214	8.7062	0.0062	−0.0002	0.0370
2	366	17.6436	9.4093	16.5298	1.0978	0.3241	8.0241	8.6983	0.0103	0.0009	0.0538
3	402	17.2985	9.3203	17.0511	1.0803	0.2790	8.0544	8.7170	0.0089	−0.0001	0.1161
4	400	17.0484	8.8683	16.8791	1.1282	0.3046	8.0476	8.7265	0.0088	0.0002	0.1293
5	385	16.6422	8.7327	16.5241	1.0930	0.3248	8.0391	8.7149	0.0099	0.0005	0.1224
6	330	16.0602	8.4877	16.1892	1.0878	0.2836	8.0298	8.6983	0.0101	0.0010	0.1015
7	328	15.7817	8.3099	17.0209	1.1012	0.2936	8.0607	8.7128	0.0091	−0.0001	0.0777
8	301	16.1197	8.6744	17.0018	1.0982	0.2772	8.0716	8.7274	0.0110	0.0010	0.1413
9	283	15.7957	8.3085	16.2070	1.1169	0.3781	8.0466	8.7303	0.0099	0.0012	0.1580
10	242	15.7498	8.3263	15.9463	1.1379	0.4466	8.0701	8.7431	0.0142	0.0012	0.1625
11	226	16.0712	8.3359	16.8212	1.1788	0.4189	8.0649	8.7559	0.0105	0.0003	0.1806
12	209	15.8975	8.3967	16.6856	1.1585	0.4190	8.0590	8.7313	0.0138	0.0013	0.2471
13	179	15.3968	7.7787	15.7669	1.1452	0.4954	8.0300	8.7377	0.0118	0.0005	0.1953
14	162	14.9709	7.6244	16.1410	1.1308	0.4795	8.0354	8.7500	0.0106	0.0010	0.2153
15	163	15.1065	7.4946	16.0238	1.1850	0.5015	7.9880	8.7484	0.0090	0.0012	0.1593
16	153	14.9202	7.4379	16.0506	1.1673	0.5212	8.0195	8.7736	0.0118	0.0011	0.1385
17	146	15.6259	7.9184	15.6923	1.1992	0.5102	8.0325	8.7632	0.0140	0.0008	0.1563
18	140	14.6176	7.6092	15.8701	0.9972	0.4340	8.0629	8.7667	0.0103	0.0007	0.1434

This table shows the univariate mean statistics of forward-looking statements, tone of CEO and other executive participants, number of words spoken by CEO and all executives in a conference call, and firm performance measured as return on assets, unexpected earnings, and annual buy-and-hold returns. A sample of firms with CEOs who were appointed after 2005 is used in the analysis. All variables are defined in Appendix B.

CEO Tenure and Forward-Looking Disclosures

Main Evidence

We begin our empirical analysis by examining the relation between CEOs' future-oriented disclosures and their tenure. We align all newly hired CEOs in event-time by their tenure-quarter and examine changes in their forward-looking disclosures over the subsequent quarters of their appointments. Because the relation between CEO tenure and their forward-looking disclosures is likely nonlinear (as suggested by the Bayesian learning model), we use both logarithmic and piecewise-linear (dummy variables for each tenure year and yearly splines) functions of tenure to estimate the empirical relation.

Table 5 provides evidence on the relation between CEO tenure and forward-looking disclosures made by CEO (*CEOFLS*) and other non-CEO officers (*NonCEOFLS*) in earnings conference calls. In Panel A, we use a longitudinal research design and model *CEOFLS* (*CEOFLSAlt*) and *NonCEOFLS* as a function of CEO tenure and other covariates as follows:

$$\begin{aligned}
 FLS = & \alpha_0 + \alpha_1 CEOTenure + \alpha_2 UE + \alpha_3 Loss + \alpha_4 UE \times Loss + \alpha_5 ROA + \alpha_6 Return + \alpha_7 Size + \alpha_8 MTB + \alpha_9 Leverage \\
 & + \alpha_{10} EarnVol + \alpha_{11} RetVol + \alpha_{12} SpecialItems + \alpha_{13} DiscOperations + \alpha_{14} \Delta CAPX + \alpha_{15} \Delta R\&D + \alpha_{16} \Delta ADV \\
 & + \alpha_{17} Following + \theta_1 Time FE + \theta_2 FirmCEO FE + \varepsilon,
 \end{aligned}
 \tag{1}$$

where *FLS* corresponds to either a CEO's or other non-CEO executives' forward-looking disclosures. The variable *CEOTenure* is the natural logarithm of the number of quarters since the CEO took office.²⁴ Following related work by Tetlock (2007), Tetlock, Saar-Tsechansky, and Macskassy (2008), Li (2010a), Price, Doran, Peterson, and Bliss (2012), Muslu et al. (2015), and Pan et al. (2015), we include observable firm characteristics that capture differences in firm information environments, operations, performance, and uncertainty. These variables include unexpected earnings (calculated relative to analyst consensus

²⁴ Logarithmically transforming CEO tenure allows us to capture a non-linear relation between tenure and disclosure style as a Taylor series expansion of logged tenure leads to higher degree polynomials (e.g., tenure, tenure²) being implicitly included in the estimation.

TABLE 5
Forward-Looking Information in Earnings Calls over CEO Tenure

Panel A: Logarithmic Transformation of CEO Tenure

	<i>CEOFLS</i> (1)	<i>CEOFLS</i> Alt (2)	<i>NonCEOFLS</i> (3)
<i>CEOTenure</i>	−0.851*** (−3.54)	−0.582*** (−3.58)	0.262 (0.86)
<i>UE</i>	−10.873 (−0.52)	−2.691 (−0.24)	22.063 (1.31)
<i>Loss</i>	0.089 (0.29)	−0.096 (−0.48)	0.356 (1.04)
<i>Loss</i> × <i>UE</i>	18.744 (0.75)	7.104 (0.50)	−9.128 (−0.34)
<i>ROA</i>	−2.076 (−0.47)	−0.028 (−0.01)	−2.742 (−0.53)
<i>Return</i>	−0.667** (−2.43)	−0.363** (−2.10)	−0.224 (−0.76)
<i>Size</i>	1.211*** (3.02)	0.593** (2.36)	0.413 (0.88)
<i>MTB</i>	−0.423 (−1.58)	−0.193 (−1.20)	0.430 (1.39)
<i>Leverage</i>	−0.591 (−0.42)	−0.276 (−0.29)	−3.608* (−1.67)
<i>EarnVol</i>	6.887 (0.94)	5.347 (1.01)	19.238** (2.07)
<i>RetVol</i>	20.840* (1.73)	2.829 (0.40)	6.279 (0.50)
<i>SpecialItems</i>	0.128 (0.47)	0.144 (0.88)	−0.171 (−0.55)
<i>DiscOperations</i>	0.623 (0.21)	−0.225 (−0.12)	2.189 (0.63)
Δ CAPX	1.083 (0.32)	−0.301 (−0.16)	−2.116 (−0.64)
Δ R&D	−2.723 (−0.63)	−1.967 (−1.07)	−0.630 (−0.14)
Δ ADV	1.750 (0.84)	1.678* (1.87)	8.466*** (5.05)
<i>Following</i>	0.008 (0.05)	−0.010 (−0.10)	−0.270 (−1.50)
Firm-CEO Fixed Effects	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes
Observations	5,581	5,581	5,581
Adjusted R ²	0.427	0.524	0.340

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This panel shows the estimated coefficients from panel regressions of the proportion of words in forward-looking statements in earnings conference calls by CEO (*CEOFLS*, *CEOFLS*Alt) and by other executives (*NonCEOFLS*) on logged CEO tenure and firm characteristics. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses. All variables are defined in Appendix B.

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forecast [*UE*]; profitability (measured by return on assets [*ROA*], loss indicator [*Loss*], annual return [*Return*]); firm growth options (measured by market-to-book [*MTB*]); operating environment (measured by firm size [*Size*], leverage [*Leverage*], earnings volatility [*EarnVol*], return volatility [*RetVol*], magnitude of special items [*SpecialItems*] and discontinued operations [*DiscOperations*], change in capital expenditures [Δ CAPX], changes in R&D [Δ R&D] and advertising expenses [Δ ADV], and

TABLE 5 (continued)

Panel B: Yearly Indicators and Spline Transformation of CEO Tenure

	Tenure Year Dummies (1)	Spline Specification (2)
<i>Year2</i>	−0.916*** (−3.32)	
<i>Year3</i>	−0.985** (−2.39)	
<i>Year4</i>	−1.570*** (−2.86)	
<i>Year5 and After</i>	−1.620** (−2.05)	
<i>Spline (Year1)</i>		−1.245** (−2.29)
<i>Spline (Year2)</i>		−1.315*** (−2.86)
<i>Spline (Year3)</i>		−0.392 (−0.81)
<i>Spline (Year4)</i>		−1.108** (−2.41)
<i>Spline (Year5)</i>		−0.631 (−0.91)
Controls	Yes	Yes
Firm-CEO Fixed Effects	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes
Observations	5,581	5,581
Adjusted R ²	0.427	0.428

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This panel shows the estimated coefficients from panel regressions of the proportion of words in forward-looking statements in earnings conference calls by CEO (*CEOFLS*) on CEO tenure year dummies (Column (1)) and splines that correspond to CEO tenure years (Column (2)). *Year2*, *Year3*, and *Year4* are indicator variables that equal 1 if an observation corresponds to the second, third, and fourth year of CEO tenure, respectively, and 0 otherwise. *Year5 and After* equals 1 if the observation corresponds to the fifth or later year of CEO tenure, and 0 otherwise. *Spline (Year i)* is the spline for the four quarters in the *i*th year of CEO appointment. Following Pan et al. (2015), *Spline (Year1)* takes the value of 1/4 to 1 for the first year of CEO tenure, while the remaining splines take the value of 0. For the second year of CEO tenure, *Spline (Year1)* takes the value of 1, *Spline (Year2)* takes the value of 1/4 to 1, while the remaining splines take the value of 0. *Spline (Year3)*, *Spline (Year4)*, and *Spline (Year5)* are defined in a similar manner. As in Panel A, control variables, firm-CEO fixed effects, year-quarter fixed effects, and a constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. *t*-statistics are in parentheses.

(continued on next page)

analyst following [Following]).²⁵ In all specifications, we include calendar year-quarter fixed effects intended to account for market-wide factors that could affect firm-level voluntary disclosures. We also estimate Equation (1) with firm-CEO fixed effects that identify any time-invariant differences across firm-CEO pairs.

If CEOs' forward-looking disclosures change with their tenure, then we expect the coefficient estimate for *CEOTenure*, α_1 , to be significant when *CEOFLS* is used as the dependent variable. We find a significant and economically meaningful negative association between CEO tenure and *CEOFLS*, controlling for relevant firm characteristics, and including firm-CEO and time fixed effects. The regression estimates suggest that after the first year of tenure, the percentage of words in forward-looking statements decreases by 1.369 percent ($= 0.851 \times \ln(5/1)$). Given that the average *CEOFLS* in the first tenure-quarter is 17.957 percent (see Table 4), the 1.369 percent decrease in *CEOFLS* in the first year is equivalent to about a 7.6 percent ($= 1.369/17.957$) relative drop in the amount of CEO forward-looking information. Similarly, after the first four years of tenure, the proportion of words in forward-looking statements by a CEO decreases by 2.411 percent ($= 0.851 \times \ln(17/1)$), which corresponds to 13.43 percent ($= 2.411/17.957$) relative reduction in CEO forward-looking information over the first four years

²⁵ We do not include CEO age as a variable in our analyses because with firm-CEO fixed effects included, CEO age is identical to CEO tenure.

TABLE 5 (continued)

Panel C: Average Values in each CEO Tenure-Quarter

Tenure-Quarters	Observations	CEOFLS ^a	NonCEOFLS ^a
1	288	0.7840	-0.1812
2	366	0.8305	0.1083
3	402	0.5921	0.2304
4	400	0.4393	0.1456
5	385	0.4524	0.07582
6	330	-0.2642	-0.3623
7	328	-0.6049	0.2621
8	301	-0.0138	0.3236
9	283	0.0427	0.0110
10	242	-0.0054	-0.3582
11	226	-0.2325	0.2477
12	209	-0.1230	0.3218
13	179	-0.2995	-0.3268
14	162	-0.6899	0.0450
15	163	-0.6251	-0.0693
16	153	-0.6055	-0.0421
17	146	0.0194	-0.0018
18	140	-1.2145	0.0676

^a *CEOFLS* and *NonCEOFLS* are orthogonalized with respect to firm performance, annual stock return, unexpected earnings, size, market-to-book, leverage, earnings and return volatility, special items, discontinued operations, change in capital expenditures scaled by total assets, change in R&D expense scaled by total assets, change in advertising expense scaled by total assets, analyst following, and firm-CEO and year-quarter fixed effects. A sample of firms with CEOs who were appointed after 2005 is used in the analysis. This panel shows the average values of the orthogonalized *CEOFLS* and *NonCEOFLS* in each of the first 18 CEO tenure-quarters.

of tenure. We observe similar results when we use *CEOFLS*_{Alt} as the dependent variable (Column (2) in Table 5, Panel A). Further, we find no (statistically or economically) significant relation between the proportion of words in non-CEO officers' forward-looking statements and their CEO's tenure (Panel A, Column (3)), suggesting that the observed effect is only CEO-specific. Not observing a relation between non-CEOs' disclosure styles and their CEO's tenure does not imply that CEOs and non-CEOs disagree or contradict each other on the conference call. It only implies that CEO tenure is not a determinant of non-CEOs' disclosure styles. In fact, the correlation between CEOs' and non-CEOs' forward-looking disclosures is 24.5 percent ($p < 0.01$), suggesting agreement in executives' disclosures.

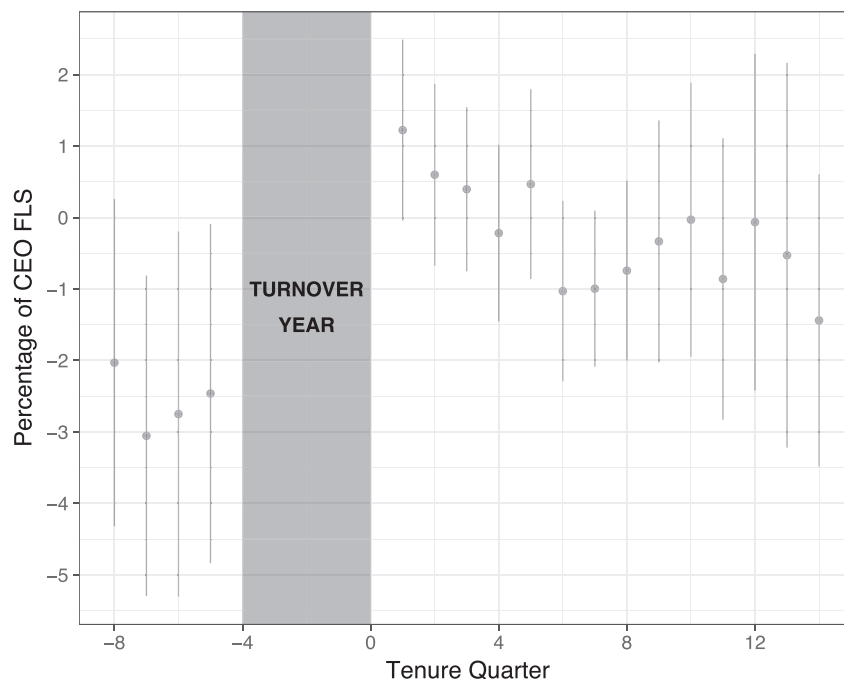
Panel B of Table 5 reports coefficient estimates for piece-wise linear specifications of CEO tenure in Equation (1). First, we create indicator variables for the first five years of CEO tenure and estimate a version of Equation (1) using these yearly indicators. This enables us to capture changes in *CEOFLS* in the first five years of CEO tenure (see Column (1)). We find that CEOs reduce disclosures of forward-looking information over time, as indicated by significant negative and increasing in magnitude coefficient estimates for yearly tenure indicators. Second, following Friedman (1991) and Pan et al. (2015), we estimate a spline specification of Equation (1) with cutoff points at the first, second, third, fourth, and fifth CEO tenure years (see Column (2)). This specification allows us to separately estimate the slope of the relation between CEO tenure and forward-looking disclosures for each of the first five years of CEO tenure. We find that the slope estimates are negative, although not always significant. In terms of economic magnitudes, the spline regression suggests that the percentage of forward-looking information by CEO after the first year of tenure decreases by 1.254 percent, which is equivalent to a 7 percent ($= 1.245/17.957$) relative decrease in the amount of forward-looking information disclosed by the CEO after the first year of tenure.²⁶ Similarly, after the first four years of tenure, we document that *CEOFLS* decreases by 4.06 percent ($= 1.245 + 1.315 + 0.392 + 1.108$), which is equivalent to a 22.6 percent ($= 4.06/17.957$) relative drop in the amount of CEO forward-looking information after the first four years of tenure.

In Panel C of Table 5, we report conditional means of *CEOFLS* and *NonCEOFLS* for the first 18 tenure-quarters for CEOs in our sample. Specifically, to account for firm characteristics, time-invariant firm-CEO characteristics, and time trends that are likely correlated with forward-looking disclosures, we orthogonalize *CEOFLS* and *NonCEOFLS* with respect to variables in Equation (1) (excluding *CEOTenure*). Consequently, our tabulated results display the portion of forward-looking information

²⁶ According to Table 4, the average *CEOFLS* in the first tenure-quarter is 17.957 percent.

FIGURE 2
Disclosure Styles of Departing and Newly Appointed CEOs over Tenure

Panel A: CEO FLS



(continued on next page)

that is not explained by firm characteristics and unobservable firm-CEO and time fixed effects. Similar to our results in Panels A and B, we observe a declining trend in the orthogonalized *CEOFLS* over *CEOTenure*. The difference between the first and 18th quarters, 1.99, is equal to 13 percent (30.7 percent) of the median (standard deviation) *CEOFLS*. Also, we find no such decline in forward-looking disclosures by non-CEO officers (*NonCEOFLS*).

Figure 1, Panel A plots point averages of fitted values of CEO-specific forward-looking statements. Fitted values of *CEOFLS* are estimated using spline regressions in Panel B of Table 5. Consistent with the patterns in Table 5, we observe a declining trend in the proportion of forward-looking disclosures by CEOs. In Figure 2, Panel A, we also plot point averages of *CEOFLS*, orthogonalized to firm characteristics and firm-CEO and time fixed effects, for pairs of departing and newly appointed CEOs.^{27,28} There is a discernible discontinuity in *CEOFLS* around CEO turnover years, with departing CEOs providing less forward-looking information than their successors. These trends are consistent with the earlier documented dynamics of CEO disclosures over CEO tenure.

Overall, our findings that CEOs' forward-looking disclosures in earnings conference calls decline in their tenure are consistent with the reduction in uncertainty about CEO ability to manage firms' assets to create value. In the early years of CEO tenure, there is greater uncertainty about CEO ability and vision for the firm. Consequently, a CEO is likely to respond to investors' demand for information to resolve this uncertainty. Over time, investors' beliefs about their management's ability become increasingly precise, leading to a lower demand for and supply of uncertainty-resolving information.²⁹

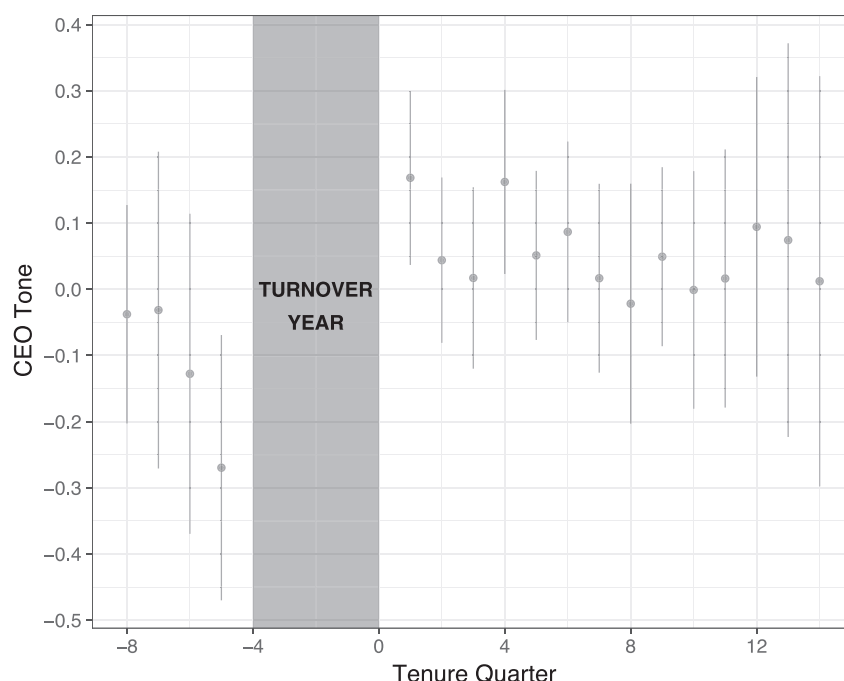
²⁷ To select pairs of departing and newly appointed CEOs with sufficient data, we identify CEO turnovers in the sample with at least two years of tenure for departing CEOs and at least three years of tenure for new CEOs. This requirement reduces our sample to 1,511 observations corresponding to 183 turnovers.

²⁸ We plot averages of orthogonalized *CEOFLS*, rather than fitted values from spline regressions, since it is not feasible to run a spline regression on a subsample of departing CEOs.

²⁹ There may be other changes within the firm (e.g., restructuring, reorganization, mergers and acquisitions [M&A]) that heighten investors' uncertainty about the firm, resulting in a greater demand for information. However, in our study, we focus on the uncertainty about the ability of new leadership (CEO-specific uncertainty), which declines over a CEO's tenure. In all our tests, we include idiosyncratic return volatility to control for the general firm-level uncertainty. In untabulated analyses, we also include indicator variables for restructuring, acquisitions, restatements, write-offs, and fraud announcements in each quarter, and our inferences regarding CEO tenure and forward-looking disclosures remain the same.

FIGURE 2 (continued)

Panel B: CEO Tone



This figure plots point averages of CEO-specific forward-looking statements (*CEOFLS*) and CEO-specific tone (*CEOTone*) in earnings conference calls for 183 pairs of departing and newly appointed CEOs. A subsample of departing and newly appointed CEOs was constructed by identifying all CEO turnovers in the sample and requiring a minimum of two years of tenure for a departing CEO. The figure plots *CEOFLS* and *CEOTone* for the last two years of appointment for departing CEOs and the first three years of appointment for new CEOs. *CEOFLS* and *CEOTone* are orthogonalized with respect to firm performance, annual stock return, unexpected earnings, size, market-to-book, leverage, earnings and return volatility, special items, discontinued operations, capital expenditures, R&D expense, advertising expense, analyst following, and year-quarter fixed effects. Panel A shows averages of orthogonalized *CEOFLS*. Panel B shows averages of orthogonalized *CEOTone*.

All variables are defined in Appendix B.

Forward-Looking Disclosures and Prior Uncertainty about CEO Ability

Even though we examine the dynamic or time-varying nature of CEO disclosures, investor information demand potentially differs cross-sectionally due to time-invariant managerial characteristics, such as whether a CEO is hired from outside the company, CEO's prior executive experience in public companies, and CEO's age at appointment. Consistent with Pan et al. (2015), we posit that CEOs who are externally hired, lack prior executive experience, and are younger have greater *ex ante* uncertainty about their abilities to create value for their firms. Accordingly, we examine whether prior uncertainty about CEO ability affects CEOs' forward-looking disclosures.

We use BoardEx to gather background information on CEOs in our sample. We create employment indicator variables: CEOs with no prior employment information in their company are classified as external hires (*ExtHire*); CEOs with no prior CEO, CFO, or COO experience in any other U.S. public company are classified as CEOs with no prior executive experience (*NoExecExp*); finally, CEOs appointed at age 52 or younger (median age of CEO at appointment in our sample) are classified as young CEOs (*YoungCEO*). We include each indicator variable and its interaction with *CEOTenure* in our model specification of Equation (1). If CEOs with greater ability uncertainty provide more forward-looking disclosures to mitigate such uncertainty, then we expect positive and significant coefficients on *ExtHire*, *NoExecExp*, and *YoungCEO* when estimating Equation (1).

Table 6 reports our results. In all specifications, we include measures related to firm characteristics (as in Tables 5 and 7) and industry and year-quarter fixed effects. In Columns (1) and (2), the coefficient estimates on *ExtHire* and *NoExecExp* are 2.570 ($p < 0.01$) and 1.644 ($p < 0.05$), respectively, suggesting that externally hired CEOs and those lacking prior executive experience provide more forward-looking disclosures than their internally hired and experienced counterparts. Further, the interaction terms between *CEOTenure* and *ExtHire* and *CEOTenure* and *NoExecExp* are negative and significant, implying that

TABLE 6
CEO Forward-Looking Disclosures by Hire Type, Experience, and Age

	<i>CEOFLS</i>		
	(1)	(2)	(3)
<i>CEOTenure</i>	−0.609*** (−3.60)	−0.701*** (−4.18)	−0.833*** (−4.28)
<i>ExtHire</i>	2.570*** (3.94)		
<i>NoExecExp</i>		1.644** (2.40)	
<i>YoungCEO</i>			−0.458 (−0.79)
<i>ExtHire_Tenure</i>	−0.900*** (−3.49)		
<i>NoExecExp_Tenure</i>		−0.621** (−2.15)	
<i>YoungCEO_Tenure</i>			−0.055 (−0.24)
Controls	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes
Observations	5,563	5,563	5,563
Adjusted R-	0.073	0.064	0.063

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from panel regressions of the amount of CEO-specific forward-looking disclosures (*CEOFLS*) in earnings conference calls on CEO tenure, CEO hire type (external or internal) and its interaction with CEO tenure (Column (1)), prior executive experience (yes or no) and its interaction with CEO tenure (Column (2)), age (below or above the median appointment age) and its interaction with CEO tenure (Column (3)), and other control variables used in Table 5. Industry fixed effects (using Fama-French 48 industry classification), year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses. All variables are defined in Appendix B.

the negative relation between forward-looking disclosures and tenure is amplified for externally hired CEOs and those lacking prior executive experience. We do not find any relation between *YoungCEO* and forward-looking disclosures. Overall, our cross-sectional patterns for two out of the three ability uncertainty measures suggest that CEOs provide more forward-looking disclosures when uncertainty about their ability is higher.

CEO Tenure and Optimism in Earnings Calls

Main Evidence

Even though forward-looking information accompanied by financial statements informs external stakeholders about their firm's future prospects, there are other attributes of disclosures, such as linguistic tone (positive or negative), that potentially influence the perceptions of external constituencies regarding firm performance and prospects (e.g., [Feldman et al. 2010](#); [Li 2010a](#); [Loughran and McDonald 2011](#)). Unlike forward-looking statements that measure “what” information CEOs disclose, the tone of CEOs' statements measures “how” they present it (i.e., CEO optimism). Therefore, in this section, we focus on how CEOs' optimism in earnings conference calls changes over the course of their tenure.

Table 7 reports the results. In Panel A, we use a longitudinal research design and model *CEOTone* and *NonCEOTone* as a function of logged *CEOTenure* and relevant covariates. More specifically, we estimate the following model:

$$\begin{aligned}
 \text{Tone} = & \alpha_0 + \alpha_1 \text{CEOTenure} + \alpha_2 \text{UE} + \alpha_3 \text{Loss} + \alpha_4 \text{UE} \times \text{Loss} + \alpha_5 \text{ROA} + \alpha_6 \text{Return} + \alpha_7 \text{Size} + \alpha_8 \text{MTB} + \alpha_9 \text{Leverage} \\
 & + \alpha_{10} \text{EarnVol} + \alpha_{11} \text{RetVol} + \alpha_{12} \text{SpecialItems} + \alpha_{13} \text{DiscOperations} + \alpha_{14} \Delta \text{CAPX} + \alpha_{15} \Delta \text{R\&D} + \alpha_{16} \Delta \text{ADV} \\
 & + \alpha_{17} \text{Following} + \theta_1 \text{Time FE} + \theta_2 \text{FirmCEO FE} + \varepsilon,
 \end{aligned}
 \tag{2}$$

where *Tone* corresponds to either the CEO's or other non-CEO executives' linguistic tone. We calculate *Tone* as the difference

TABLE 7
Tone of Earnings Calls over CEO Tenure

Panel A: Logarithmic Transformation of CEO Tenure

	<i>CEOTone</i> (1)	<i>NonCEOTone</i> (2)
<i>CEOTenure</i>	−0.099*** (−3.18)	−0.051 (−1.60)
<i>UE</i>	10.086*** (4.53)	10.615*** (4.96)
<i>Loss</i>	−0.038 (−1.06)	−0.108*** (−3.42)
<i>Loss</i> × <i>UE</i>	−5.613** (−2.05)	−5.849** (−2.02)
<i>ROA</i>	0.312 (0.68)	1.709*** (3.89)
<i>Return</i>	0.188*** (6.08)	0.180*** (5.38)
<i>Size</i>	−0.146** (−2.56)	−0.146*** (−2.60)
<i>MTB</i>	0.004 (0.12)	0.041 (1.28)
<i>Leverage</i>	0.095 (0.51)	0.087 (0.45)
<i>EarnVol</i>	0.384 (0.58)	0.840 (1.41)
<i>RetVol</i>	−3.765*** (−2.71)	−7.605*** (−5.19)
<i>SpecialItems</i>	0.079*** (3.10)	0.067** (2.29)
<i>DiscOperations</i>	0.356 (1.08)	0.130 (0.35)
Δ CAPX	−0.591* (−1.72)	−0.318 (−0.84)
Δ R&D	0.506 (1.37)	0.612 (0.84)
Δ ADV	0.694*** (4.59)	−0.123 (−0.73)
<i>Following</i>	−0.018 (−1.17)	−0.001 (−0.07)
Firm-CEO Fixed Effects	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes
Observations	5,581	5,581
Adjusted R ²	0.591	0.547

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This panel shows the estimated coefficients from a panel regression of CEO tone in earnings conference calls (*CEOTone*) and by other executives (*NonCEOTone*) on logged CEO tenure and firm characteristics. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses.

All variables are defined in Appendix B.

(continued on next page)

between the number of positive and negative words scaled by the total number words spoken by the CEO or other non-CEO firm officers in the conference call. *CEOTenure* is the natural logarithm of the number of quarters since the CEO's appointment. As in Equation (1), we include observable firm characteristics that capture cross-sectional differences in firm's information environment, operations, and performance. We also include calendar year-quarter and firm-CEO fixed effects.

TABLE 7 (continued)

Panel B: Yearly Indicators and Spline Transformation of CEO Tenure

	Tenure Year Dummies (1)	Spline Regression (2)
<i>Year2</i>	−0.087*** (−2.76)	
<i>Year3</i>	−0.131*** (−2.64)	
<i>Year4</i>	−0.169** (−2.44)	
<i>Year5 and After</i>	−0.224** (−2.28)	
<i>Spline (Year1)</i>		−0.187*** (−3.10)
<i>Spline (Year2)</i>		−0.098* (−1.91)
<i>Spline (Year3)</i>		−0.074 (−1.37)
<i>Spline (Year4)</i>		−0.089 (−1.50)
<i>Spline (Year5)</i>		−0.104 (−1.37)
Controls	Yes	Yes
Firm-CEO Fixed Effects	Yes	Yes
Year-Qtr Fixed Effects	Yes	Yes
Observations	5,581	5,581
Adjusted R ²	0.590	0.591

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This panel shows the estimated coefficients from panel regressions of CEO tone in earnings conference calls (*CEOTone*) on CEO tenure year dummies (Column (1)) and splines that correspond to CEO tenure years (Column (2)). *Year2*, *Year3*, and *Year4* are indicator variables that equal 1 if an observation corresponds to the second, third, and fourth year of CEO tenure, respectively, and 0 otherwise. *Year5 and After* equals 1 if the observation corresponds to the fifth or later year of CEO tenure, and 0 otherwise. *Spline (Year i)* is the spline for the four quarters in the i th year of CEO appointment. Following Pan et al. (2015), *Spline (Year1)* takes the value of 1/4 to 1 for the first year of CEO tenure, while the remaining splines take the value of 0. For the second year of CEO tenure, *Spline (Year1)* takes the value of 1, *Spline (Year2)* takes the value of 1/4 to 1, while the remaining splines take the value of 0. *Spline (Year3)*, *Spline (Year4)*, and *Spline (Year5)* are defined in a similar manner. As in Panel A, control variables, firm-CEO fixed effects, year-quarter fixed effects, and a constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses.

(continued on next page)

If CEOs' optimism (as measured by linguistic tone) declines (increases) in their tenure, then we expect the coefficient estimate for *CEOTenure*, β_1 , to be negative (positive) and significant when *CEOTone* is the dependent variable. We find a significant and economically meaningful negative association between CEO tenure and CEO optimism, while including firm characteristics, firm-CEO, and time fixed effects. After the first year of CEO tenure, we find a 0.159 ($= 0.099 \times \ln(5/1)$) decrease in *CEOTone*. Given that the average *CEOTone* in the first tenure-quarter is 1.168 (as per Table 4), the 0.159 decrease in *CEOTone* translates to a 13.61 percent ($= 0.159/1.168$) relative decrease in CEO tone after the first year of tenure. Similarly, after the first four years of CEO tenure, there is a 0.280 ($= 0.099 \times \ln(17/1)$) decrease in *CEOTone*, corresponding to a 23.97 percent ($= 0.280/1.168$) relative reduction in *CEOTone* over the first four years of tenure. We find no such decline for non-CEO tone. The correlation between CEO and non-CEO tone is 46 percent ($p < 0.01$), suggesting that while *CEOTenure* is not a common determinant of optimism across executives, there is a strong agreement in optimism between the CEO and other executives on a conference call.

Panel B of Table 7 reports coefficient estimates for piece-wise linear specifications of *CEOTenure* in Equation (2). In Column (1), we create indicator variables for the first five years of CEO tenure and estimate a version of Equation (2) with these yearly indicators. In Column (2), we estimate a spline specification of Equation (2) with cutoff points at the first, second, third, fourth, and fifth CEO tenure year. These specifications allow us to separately analyze the relation between CEO tenure and

TABLE 7 (continued)

Panel C: Average Values in each CEO Tenure-Quarter

Tenure-Quarters	Observations	<i>CEOTone</i> ^a	<i>NonCEOTone</i> ^a
1	288	0.1071	0.0068
2	366	0.0095	0.0198
3	402	0.0137	-0.0225
4	400	0.0267	-0.0027
5	385	-0.0140	-0.0090
6	330	-0.0138	-0.0386
7	328	0.0026	-0.0332
8	301	-0.0014	-0.0731
9	283	-0.0101	0.0142
10	242	-0.0226	0.0553
11	226	0.0082	0.0190
12	209	-0.0192	-0.0149
13	179	-0.0142	0.0456
14	162	-0.0382	0.0163
15	163	0.0228	0.0120
16	153	-0.0394	0.0111
17	146	0.0054	0.0033
18	140	-0.1408	-0.0427

^a *CEOTone* and *NonCEOTone* are orthogonalized with respect to firm performance, annual stock return, unexpected earnings, size, market-to-book, leverage, earnings and return volatility, special items, discontinued operations, change in R&D expense scaled by total assets, change in advertising expense scaled by total assets, analyst following, and firm-CEO and year-quarter fixed effects. A sample of firms with CEOs who were appointed after 2005 is used in the analysis.

This panel shows the summary statistics of the orthogonalized *CEOTone* and *NonCEOTone* in each of the first 18 CEO tenure-quarters.

CEO optimism for each of the first five years of tenure. Consistent with our results in Panel A, we find that CEOs become less optimistic over time, as indicated by significant negative and increasing in magnitude coefficient estimates for yearly tenure indicators. Further, using the spline specification, we find that the slope estimates of CEO tenure and CEO optimism relation are negative, although not always significant. Specifically, we find that tone decreases by 0.187 after the first year of tenure, which is equivalent to a 16.01 percent ($= 0.187/1.168$) relative decrease in *CEOTone* after the first year of tenure.³⁰ Similarly, we find a 0.448 ($= 0.187 + 0.098 + 0.074 + 0.089$) decrease in *CEOTone* after the first four years of tenure, which translates to a 38.35 percent ($= 0.448/1.168$) relative reduction in *CEOTone* after the first four years of tenure.

In Panel C of Table 7, we report conditional means of *CEOTone* and *NonCEOTone* for the first 18 tenure-quarters for CEOs in our sample. Similar to our prior tests examining forward-looking disclosures, we orthogonalize the linguistic tone measures with respect to covariates in Equation (2) (excluding *CEOTenure*). As before, we document a declining pattern in the orthogonalized CEO tone over tenure. *CEOTone* declines from 0.1071 in the first tenure-quarter to 0.0267 in the fourth tenure-quarter, and continues to decline to -0.1408 in the 18th quarter. The difference between the first and 18th quarters, 0.248, is equal to 22.1 percent (31.4 percent) of the median (standard deviation) *CEOTone*. Additionally, we do not find any discernible decline in *NonCEOTone* over *CEOTenure*.³¹ In Figure 1, Panel B, we plot point averages of fitted values of CEO-specific tone over the first 18 CEO tenure-quarters. Consistent with our results in Table 7, we observe a declining trend in CEOs' linguistic tone over their tenure. Further, in Figure 2, Panel B, we plot point averages of *CEOTone*, orthogonalized to firm characteristics and firm-CEO and time fixed effects, for pairs of departing and newly appointed CEOs. There is a pronounced discontinuity in *CEOTone* around CEO turnover years—departing CEOs are less optimistic in their statements than their successors. These trends are consistent with the previously documented dynamics of CEO disclosure tone over CEO tenure.

³⁰ In Table 4, the average value of *CEOTone* in the first tenure-quarter is 1.168.

³¹ In untabulated analyses, we examine patterns in firm performance over CEO tenure. Specifically, we orthogonalize firms' return on assets, unexpected earnings, and stock return with respect to firm-CEO and time fixed effects and examine the changes in unexplained performance over the first 18 tenure-quarters for CEOs in our sample. We do not observe any meaningful trends in firm performance over CEO tenure, suggesting that the decline in tone over CEO tenure is not due to declining operating performance.

Overall, our finding that CEOs' optimism in earnings conference calls declines in their tenure is consistent with the attenuation of CEOs' career concerns over time. Appointment of a new CEO is accompanied by significant outsiders' uncertainty about the new leadership. This uncertainty potentially exacerbates CEOs' career concerns in the early years of their tenure, leading them to use positive (or eschew negative) linguistic tone to favorably influence external parties. Over the course of their tenure, as this uncertainty is resolved, CEOs' career concerns diminish, attenuating their incentives to favorably influence outsiders' beliefs through disclosures.

Market Perceptions of CEO Optimism Over the Course of Tenure

Since our findings suggest that CEOs are more optimistic earlier in their tenure, and prior studies document a positive relation between the tone of earnings conference calls and event-period abnormal returns (e.g., Miller 2010; Price et al. 2012), we examine how market reactions to CEO tone change with CEO tenure. Holmstrom (1999) posits that managerial career concerns motivate CEOs to manipulate their performance (i.e., signal-jamming) to favorably influence outsiders' perceptions, because their ability is unknown and can only be ascertained from periodic reports. In equilibrium, however, investors correctly adjust for the inflated performance (Stein 1989). In spite of being unable to "fool" the market, managers continue to manipulate performance signals (Stein 1989).³² Nevertheless, managers' incentives to do so decrease over time as the uncertainty about their ability decreases. In our context, we document that due to career concerns, CEOs attempt to influence outsiders' perceptions of their ability by using a more optimistic tone in earnings conference calls early in their tenure. Following Stein (1989) and Holmstrom (1999), we predict that the market adjusts for this "manipulation" by discounting excessive CEO optimism in early years of CEO tenure.

Table 8 shows the results of a three-day event-period abnormal returns ($BHAR[0, 2]$) regressed on $CEOTone$, an indicator variable for early CEO tenure ($EarlyTenure$), their interaction ($CEOTone \times EarlyTenure$), and other controls. The negative coefficient on the interaction term, $CEOTone \times EarlyTenure$, of -0.007 ($p < 0.05$) compared to the positive coefficient on $CEOTone$ of 0.015 ($p < 0.01$) suggests that the market reaction to CEO tone is muted by approximately half in the early periods of CEO tenure. In untabulated tests, we find similar inferences when we estimate the relationship between abnormal returns and CEO tone on CEO tenure tercile subsamples. These findings are consistent with theoretical models in Stein (1989) and Holmstrom (1999), and suggest that the market rationally discounts CEOs' excessive optimism early in their tenure.

CEO Optimism and Heightened Career Concerns

To provide further evidence on the CEO optimism-tenure relation, we examine differences in disclosure tone among CEOs with differing career concerns. We use CEO hiring type (external versus internal hire), CEOs' prior executive experience in public companies, and CEOs' age at appointment as our measures of CEO career concerns.³³ If CEOs with greater career concerns use more optimistic language to favorably influence the market's beliefs of their ability, then we expect positive and significant coefficients on $ExtHire$, $NoExecExp$, and $YoungCEO$ when estimating Equation (2).

Table 9 reports our results. In Columns (1) and (2), we find no relation between $ExtHire$ and $NoExecExp$ and CEO disclosure optimism. In Column (3), $YoungCEO$ is equal to 1 if CEO was appointed at age 52 (median CEO appointment age in our sample) or younger, and 0 otherwise. We find that $YoungCEO$ is positive and significant (0.137 , $p < 0.05$), indicating that younger CEOs are, on average, more optimistic than their older counterparts. However, there is no significant difference in the decline of CEO optimism over tenure for young CEOs (coefficient $YoungCEO_Tenure$ is insignificant). Given that we do not observe consistent results for all three measures of CEO career concerns, it is unclear to what extent these CEO characteristics influence CEO disclosure optimism.

Dynamics of Positive and Negative CEO Tone

While our results in Table 7 suggest a significant decline in CEOs' overall tone over their tenure, it remains unclear whether, over time, CEOs use less positive or more negative language or both. To further examine how CEOs communicate in earnings calls, we estimate Equation (2) using positive $CEOTone$ (i.e., the number of positive words spoken by the CEO divided by the total number of CEO words) and negative $CEOTone$ (i.e., the number of negative words spoken by the CEO

³² Although the preferred outcome is no manipulation on the part of a manager and no market unraveling of such manipulation, it is not a sustainable Nash equilibrium. If there is no market unraveling, then the manager has an incentive to manipulate the performance signal. Consequently, the manager is "forced" to inflate performance because the market always assumes signal-jamming.

³³ We note, however, that prior studies (e.g., Gibbons and Murphy 1992b; Chevalier and Ellison 1999; Holmstrom 1999) use CEO's appointment age as the main measure of CEO career concerns, as younger CEOs have longer careers ahead of them and are, thus, more concerned with their reputation, which is affected by their policies, disclosures, and subsequent performance.

TABLE 8
Event-Period Abnormal Returns, CEO Tone, and CEO Tenure

	<i>BHAR</i> [0, 2]
<i>CEOTone</i>	0.015*** (6.52)
<i>EarlyTenure</i>	0.011 (1.46)
<i>Tone</i> × <i>EarlyTenure</i>	−0.007** (−2.47)
<i>UE Decile</i>	0.004 (1.05)
<i>Loss</i>	−0.016* (−1.72)
<i>ROA</i>	−0.172 (−1.02)
<i>Return</i>	−0.028*** (−3.62)
<i>Size</i>	−0.010 (−1.33)
<i>MTB</i>	−0.007 (−1.28)
<i>Leverage</i>	−0.007 (−0.28)
<i>EarnVol</i>	−0.014 (−0.09)
<i>RetVol</i>	−0.654** (−2.05)
<i>SpecialItems</i>	0.001 (0.10)
<i>DiscOperations</i>	0.077 (0.65)
Δ CAPX	−0.067 (−0.81)
Δ R&D	0.023 (0.14)
Δ ADV	0.144*** (2.87)
<i>Following</i>	0.001 (0.21)
Firm-CEO Fixed Effects	Yes
Year-Qtr. Fixed Effects	Yes
Observations	5,522
Adjusted R ²	0.158

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from a panel regression of the three-day buy-and-hold abnormal return (*BHAR*[0, 2]) on CEO tone in earnings conference calls (*CEOTone*), CEO early tenure dummy (*EarlyTenure*), their interaction (*CEOTone* × *EarlyTenure*), firm characteristics, and year-quarter fixed effects. *BHAR*[0,2] is the cumulative abnormal return over the three-day event window calculated relative to the Fama-French three-factor benchmark, where day 0 is the conference call date. Variable *EarlyTenure* equals 1 if CEO tenure is less than or equal to six quarters (i.e., corresponds to the first tercile of CEO tenure distribution in our sample). Variable *UE Decile* is the deciles of unexpected earnings variable (*UE*), created using quarterly independent sorts of earnings conference calls by the corresponding unexpected earnings (*UE*). Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but not reported. Interactions of all control variables with *UE Decile* are included, but these interactions are not reported. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. Reported statistics are based on robust standard errors and clustering at the firm level. t-statistics are in parentheses.

All other variables are defined in Appendix B.

TABLE 9
CEO Tone and CEO Tenure by Hire Type, Experience, and Age

	<i>CEOTone</i>		
	(1)	(2)	(3)
<i>CEOTenure</i>	−0.070*** (−3.26)	−0.056*** (−2.67)	−0.054** (−2.17)
<i>ExtHire</i>	0.088 (1.14)		
<i>NoExecExp</i>		0.082 (1.11)	
<i>YoungCEO</i>			0.137** (2.00)
<i>ExtHire_Tenure</i>	−0.015 (−0.50)		
<i>NoExecExp_Tenure</i>		−0.052* (−1.70)	
<i>YoungCEO_Tenure</i>			−0.033 (−1.08)
Controls	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes
Observations	5,563	5,563	5,563
Adjusted R ²	0.124	0.120	0.119

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from panel regressions of CEO-specific tone (*CEOTone*) in earnings conference calls on CEO tenure, CEO hire type (external or internal) and its interaction with CEO tenure (Column (1)), prior executive experience (yes or no) and its interaction with CEO tenure (Column (2)), age (below or above the median appointment age) and its interaction with CEO tenure (Column (3)), and other control variables used in Table 7. Industry fixed effects (using Fama-French 48 industry classification), year-quarter fixed effects, and the constant are included in each regression, but are not reported. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. Reported statistics are based on robust standard errors and clustering at the firm level. t-statistics are in parentheses.

All variables are defined in Appendix B.

divided by the total number of CEO words) separately as dependent variables. Table 10 reports our results. We find that positive *CEOTone* is significantly negatively associated with *CEOTenure* (coefficient estimate -0.06 , $p < 0.05$), while negative *CEOTone* is significantly positively associated with *CEOTenure* (coefficient estimate 0.031 , $p < 0.05$). Figure 3 plots linear approximations of orthogonalized CEO positive tone and CEO negative tone for the first 18 tenure-quarters. Consistent with our results in Table 10, we find that over time, CEOs use less positive and more negative words in their disclosures, suggesting that our results in Table 7 are driven by both positive and negative components of tone. Collectively, our results suggest that CEOs are, on average, optimistic in their statements early on in their appointments, and as their career concerns ebb over time, CEOs become relatively less (more) optimistic (pessimistic) in their disclosures.

Dynamics of Forward-Looking and Non-Forward-Looking CEO Tone

CEOs' statements in earnings conference calls consist of both forward-looking information and information related to current and past performance. Since forward-looking statements are more likely to be optimistic, we examine whether the decline in overall *CEOTone* over tenure is primarily attributable to CEOs' forward-looking information or statements related to current and past performance. Following the methodology in Section II, we partition CEO statements into forward-looking and non-forward-looking and calculate CEO tone separately for each. We then estimate a version of Equation (2) with CEO forward-looking and non-forward-looking tone as dependent variables. The last two columns in Table 10 display our results. We find that the decline in CEO tone over tenure is most apparent in CEOs' non-forward-looking statements (coefficient estimate -0.099 , $p < 0.01$). The association between *CEOTone* in forward-looking statements and *CEOTenure* is negative, but not statistically significant. Overall, these results are consistent with the idea that CEOs use relatively more positive language early in their careers to positively influence outsiders' perception of their managerial performance and, thus, ability.

TABLE 10
CEO Tenure and CEO Tone in Earnings Calls by Tone Type and Statement Type

	Tone Type		Statement Type	
	Positive Tone	Negative Tone	Non-FLS	FLS
<i>CEOTenure</i>	−0.060** (−2.54)	0.031** (2.09)	−0.099*** (−2.97)	−0.071 (−1.49)
<i>UE</i>	7.174*** (4.58)	−2.453* (−1.83)	13.100*** (5.17)	1.768 (0.41)
<i>Loss</i>	0.029 (1.16)	0.066*** (2.92)	−0.049 (−1.20)	−0.029 (−0.50)
<i>Loss × UE</i>	−4.986*** (−2.82)	−0.146 (−0.09)	−7.311** (−2.33)	−0.317 (−0.06)
<i>ROA</i>	−0.102 (−0.31)	−0.476* (−1.89)	0.140 (0.29)	0.850 (1.07)
<i>Return</i>	0.102*** (4.42)	−0.087*** (−6.09)	0.213*** (6.66)	0.101* (1.93)
<i>Size</i>	−0.131*** (−2.95)	0.019 (0.73)	−0.147** (−2.47)	−0.141* (−1.85)
<i>MTB</i>	0.021 (0.91)	0.011 (0.72)	0.007 (0.20)	−0.021 (−0.49)
<i>Leverage</i>	0.028 (0.19)	−0.062 (−0.68)	0.063 (0.32)	0.035 (0.12)
<i>EarnVol</i>	−0.045 (−0.07)	−0.105 (−0.22)	0.015 (0.02)	0.920 (0.57)
<i>RetVol</i>	−1.360 (−1.32)	1.875** (2.43)	−3.886** (−2.55)	−4.304* (−1.86)
<i>SpecialItems</i>	0.039* (1.83)	−0.031** (−2.39)	0.085*** (3.21)	0.028 (0.52)
<i>DiscOperations</i>	0.110 (0.45)	−0.240 (−1.24)	0.286 (0.81)	0.934 (1.41)
<i>ΔCAPX</i>	−0.353 (−1.22)	0.234 (1.29)	−0.661* (−1.73)	−0.253 (−0.38)
<i>ΔR&D</i>	0.311 (1.08)	−0.205 (−0.93)	0.611 (1.48)	0.982* (1.95)
<i>ΔADV</i>	0.330** (2.11)	−0.387*** (−5.09)	0.680*** (3.87)	0.139 (0.51)
<i>Following</i>	−0.011 (−0.92)	0.009 (1.06)	−0.016 (−0.97)	−0.023 (−0.79)
Firm-CEO Fixed Effects	Yes	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes	Yes
Observations	5,581	5,581	5,581	5,581
Adjusted R ²	0.590	0.568	0.561	0.280

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

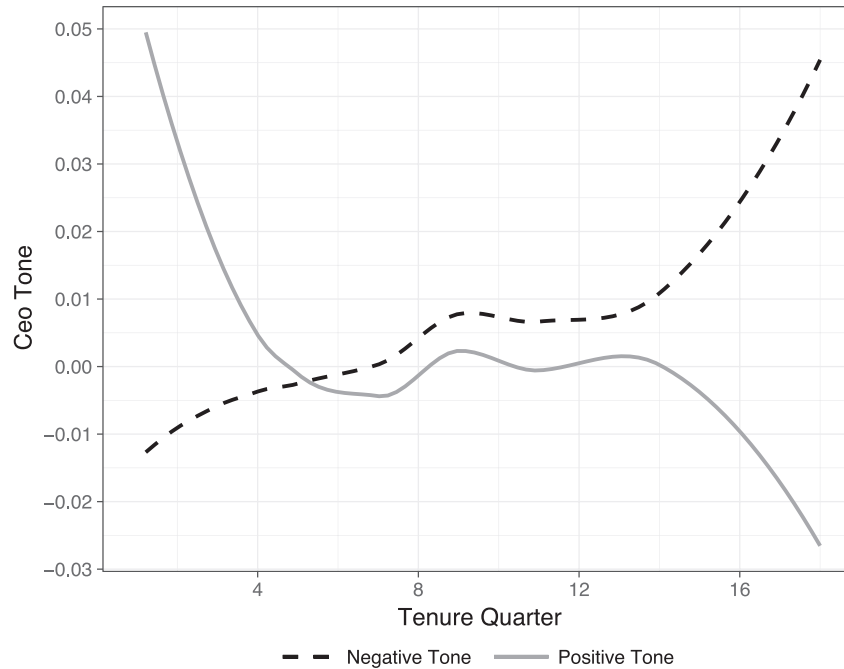
This table shows the estimated coefficients from a panel regression of the CEO-specific positive and negative tone (first two columns) and forward-looking and non-forward-looking tone (last two columns) in earnings conference calls on CEO tenure and firm characteristics. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses. All variables are defined in Appendix B.

IV. ADDITIONAL EVIDENCE AND ROBUSTNESS TESTS

CEO Appointment Following Performance- and Non-Performance-Motivated Turnovers

Appointment of a new CEO often follows poor firm performance that caused a forced turnover. These negative events potentially heighten uncertainty about firm prospects around an incoming CEO's appointment. In contrast, if a CEO is appointed following an orderly departure by the predecessor or if changes in business operations are less likely, then

FIGURE 3
CEO Tenure and CEO Positive and Negative Tone in Earnings Conference Calls



This figure plots linear approximations using the local regression (LOESS) smoothing method of CEO-specific positive tone (*Positive Tone*) and negative tone (*Negative Tone*) in earnings conference calls for their first 18 tenure-quarters. A sample of CEOs who were appointed after 2005 is used for the estimation. *Positive Tone* and *Negative Tone* measure the proportions of positive and negative words spoken by a CEO in the earnings call, respectively. *Positive Tone* and *Negative Tone* are orthogonalized with respect to firm performance, annual stock return, unexpected earnings, size, market-to-book, leverage, earnings and return volatility, special items, discontinued operations, capital expenditures, R&D expense, advertising expense, analyst following, and firm-CEO and year-quarter fixed effects.

All variables are defined in Appendix B.

uncertainty about a CEO's plans, actions, or ability is likely lower. We analyze disclosure style differences between CEOs appointed following performance-related and non-performance-related CEO turnovers to further examine whether uncertainty affects incoming CEOs' disclosure styles. We follow [Pan et al. \(2015\)](#) to identify non-performance-motivated turnovers as turnovers due to death of the departing CEO (as indicated by Execucomp), those due to retirement of the departing CEO (as indicated by Execucomp or age of the departing CEO greater than 65) at good firm performance, and turnovers that were accompanied by a positive pre-turnover industry-adjusted stock return and an increase in return on assets. We classify all other turnovers as performance-motivated.

Table 11 reports the results of estimating Equations (1) and (2) on subsamples of performance-motivated and non-performance-motivated turnovers. Consistent with our prior findings, we find that the negative association between CEO tenure and CEO forward-looking disclosures and tone persists in both subsamples, but the relation is more pronounced among CEOs appointed following a performance-related turnover. These results attest to the heightened uncertainty about new CEOs' managerial ability following performance-related CEO turnovers, affecting their disclosure styles. However, since we also observe a significant negative relation between tenure and disclosure styles of CEOs appointed following orderly departures, our results are consistent with the relations being driven by the uncertainty in the ability of all new managers, regardless of firm-level uncertainty.

Surviving CEOs

One potential caveat to our documented negative associations between CEO tenure and the amount of forward-looking information and CEO linguistic tone is that our results are potentially driven by overly optimistic or low-ability CEOs whose employment is terminated early (i.e., before the 18th tenure-quarter). To address this issue, we repeat our analysis on a subsample of "surviving" CEOs, i.e., those who remain CEO for at least 18 quarters after their appointment. Admittedly, CEOs who stay in office for all 18 quarters are likely high-ability CEOs ([Milbourn 2003](#)), but our results are unbiased since we

TABLE 11
CEO Tenure and CEO Disclosure Style
Subsamples of Performance- and Non-Performance-Motivated Turnovers

	Performance-Motivated		Non-Performance-Motivated	
	<i>CEOFLS</i>	<i>CEOTone</i>	<i>CEOFLS</i>	<i>CEOTone</i>
<i>CEOTenure</i>	−0.968*** (−3.27)	−0.103** (−2.33)	−0.758* (−1.90)	−0.083* (−1.89)
<i>UE</i>	−37.161* (−1.85)	9.300*** (3.40)	56.781 (1.16)	11.732*** (3.25)
<i>Loss</i>	−0.035 (−0.09)	−0.037 (−0.75)	0.351 (0.63)	−0.061 (−1.15)
<i>Loss × UE</i>	39.294 (1.55)	−3.780 (−1.11)	−25.379 (−0.44)	−10.857** (−2.01)
<i>ROA</i>	−3.740 (−0.73)	0.289 (0.46)	1.326 (0.17)	0.372 (0.56)
<i>Return</i>	−0.750** (−2.15)	0.195*** (4.62)	−0.547 (−1.20)	0.173*** (3.95)
<i>Size</i>	0.936* (1.82)	−0.143* (−1.93)	1.634** (2.51)	−0.157** (−1.99)
<i>MTB</i>	−0.499 (−1.42)	0.007 (0.19)	−0.308 (−0.70)	0.009 (0.19)
<i>Leverage</i>	−1.463 (−0.87)	0.157 (0.72)	1.064 (0.44)	−0.063 (−0.19)
<i>EarnVol</i>	15.560* (1.75)	0.700 (0.85)	−6.082 (−0.57)	−0.327 (−0.25)
<i>RetVol</i>	14.295 (0.99)	−4.024** (−2.34)	32.567 (1.50)	−2.858 (−1.16)
<i>SpecialItems</i>	0.378 (1.04)	0.092*** (2.77)	−0.336 (−0.81)	0.065 (1.65)
<i>DiscOperations</i>	−2.383 (−0.61)	0.331 (0.77)	5.613 (1.31)	0.449 (0.83)
<i>ΔCAPX</i>	−3.020 (−0.70)	−0.731 (−1.62)	7.317 (1.21)	−0.238 (−0.45)
<i>ΔR&D</i>	−3.260 (−0.53)	1.123*** (2.90)	0.458 (0.09)	−0.951 (−1.52)
<i>ΔADV</i>	1.700 (1.14)	0.784*** (3.19)	−18.426 (−0.82)	−3.577** (−2.13)
<i>Following</i>	0.006 (0.03)	−0.005 (−0.24)	0.077 (0.33)	−0.043* (−1.69)
Firm-CEO Fixed Effects	Yes	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes	Yes
Observations	3,229	3,229	2,352	2,352
Adjusted R ²	0.432	0.593	0.413	0.585

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from panel regressions of the amount of CEO-specific forward-looking disclosures (*CEOFLS*) and CEO-specific tone (*CEOTone*) in earnings conference calls on CEO tenure and firm characteristics. Subsamples of firms with CEOs who were appointed after 2005 as a result of performance-driven or non-performance-driven turnovers are used for the estimation. Following [Pan et al. \(2015\)](#), we identify the likely non-performance-driven turnovers as turnovers due to death of the departing CEO (as indicated by Execucomp), turnovers due to retirement of the departing CEO (as indicated by Execucomp or age of the departing CEO greater than 65) at good firm performance (when the firm's pre-turnover industry-adjusted stock return ≥ 0), and turnovers that were accompanied by a positive pre-turnover industry-adjusted stock return and positive growth in earnings scaled by total assets. The remaining turnovers are classified as performance-driven. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. t-statistics are in parentheses.

All variables are defined in Appendix B.

TABLE 12
CEO Tenure and CEO Disclosure Style
Subsample of Surviving CEOs

	<i>CEOFLS</i> (1)	<i>CEOTone</i> (2)
<i>CEOTenure</i>	−1.001*** (−2.75)	−0.106** (−2.12)
<i>UE</i>	−2.077 (−0.07)	8.926*** (2.83)
<i>Loss</i>	−0.057 (−0.15)	−0.087* (−1.67)
<i>Loss</i> × <i>UE</i>	8.875 (0.24)	−4.288 (−0.96)
<i>ROA</i>	4.018 (0.64)	−0.235 (−0.35)
<i>Return</i>	−0.861*** (−2.67)	0.215*** (5.43)
<i>Size</i>	1.017** (2.15)	−0.183*** (−2.67)
<i>MTB</i>	−0.356 (−1.23)	0.041 (1.16)
<i>Leverage</i>	−0.590 (−0.39)	0.315 (1.38)
<i>EarnVol</i>	14.391 (1.42)	0.157 (0.20)
<i>RetVol</i>	18.222 (1.16)	−5.018*** (−2.82)
<i>SpecialItems</i>	−0.147 (−0.41)	0.090** (2.60)
<i>DiscOperations</i>	−1.399 (−0.40)	0.574 (1.38)
Δ CAPX	1.725 (0.45)	−0.680* (−1.71)
Δ R&D	10.407* (1.81)	0.013 (0.01)
Δ ADV	1.603 (0.87)	0.674*** (4.77)
<i>Following</i>	0.275 (1.38)	0.002 (0.07)
Firm-CEO Fixed Effects	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes
Observations	3,355	3,355
Adjusted R ²	0.405	0.600

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from panel regressions of the amount of CEO-specific forward-looking disclosures (*CEOFLS*) and CEO-specific tone (*CEOTone*) in earnings conference calls on CEO tenure and firm characteristics. A sample of firms with CEOs who were appointed after 2005 and remained CEO for at least 18 quarters is used for the estimation. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. t-statistics are in parentheses. All variables are defined in Appendix B.

examine their disclosure styles longitudinally. Our sample of “surviving” CEOs consists of 3,355 firm-quarter observations with 223 unique CEOs. Table 12 reports the coefficient estimates for Equations (1) and (2) estimated on this reduced sample. We find that our results documenting the negative relations between the amount and tone of CEO disclosures and tenure continue to hold in this sample of “surviving” CEOs. In untabulated tests, we also examine whether there are differences in

disclosure tone and forward-looking focus among “surviving” CEOs and those that depart in the subsequent quarter. In 17 out of 18 tenure-quarters, we find no statistical differences in disclosures between the two groups of CEOs. Collectively, these findings suggest that the relation between CEOs’ disclosure styles and their tenure is not driven by low-ability CEOs using expansive disclosures and excessively positive language in the early years of their tenure.

Presentation and Questions-and-Answers Sections of Earnings Conference Calls

Results in Section III suggest that CEOs’ forward-looking disclosures and optimism in earnings conference calls decrease in CEO tenure. However, it is unclear whether CEO statements in the presentation or the questions-and-answers sections of earnings calls, or both, drive these results. The presentation section of an earnings call consists of statements that are prepared in advance and are likely strategically scripted by the CEO with assistance from legal counsel and investor relations personnel in anticipation of outsiders’ demand for information. In contrast, the questions-and-answers section of the call is spontaneous as it is prompted by call participants’ questions. If we indeed capture changes in CEOs’ disclosure styles over their tenure, our prior results should continue to hold for CEO statements made in both the strategically scripted presentation and the spontaneous questions-and-answers sections of an earnings call.

We split the text in each earnings call into presentation remarks and questions-and-answers components and then calculate *CEOFLS* and *CEOTone* separately for the two sections. Table 13 reports the coefficient estimates of Equations (1) and (2) with these modified measures as dependent variables. We find that the negative relations between CEO tenure and both forward-looking statements and linguistic tone maintain in both the prepared remarks (scripted) and questions-and-answers (unscripted) sections of the earnings call. This suggests that our results are unlikely driven by firm disclosure policies, but instead capture CEO-specific disclosures styles.

Cross-Sectional Analysis

Our prior tests use a longitudinal (panel) regression research design to examine changes in CEO disclosure style with CEO tenure. While panel models are better statistical models to study the dynamics of CEO disclosures over tenure,³⁴ cross-sectional models allow for estimation of general trends in the population while utilizing a significantly larger number of observations. We repeat our analyses in Tables 5 and 7 with all available earnings conference call observations for CEOs of U.S. public companies for the period 2006–2014. Relaxing the restriction on the CEO appointment date increases our sample size to 18,660 firm-quarter observations. Table 14 reports the coefficient estimates of Equations (1) and (2) using the cross-sectional research design. Similar to our longitudinal findings, including firm characteristics and firm-CEO and year-quarter fixed effects, we find a significant negative association between the CEOs’ forward-looking disclosures and their tenure.³⁵ We also find a strong negative association between CEOs’ relative optimism in earnings calls and their tenure. We do not observe such trends for non-CEO executives over CEO tenure. Collectively, our inference from both the longitudinal and cross-sectional designs is that CEOs’ forward-looking disclosures and optimism decline in their tenure.

V. CONCLUSION

We empirically examine the relation between CEOs’ disclosure styles (forward-looking information and optimism) and their tenure. Using a longitudinal research design on a large sample of earnings conference calls over the 2006–2014 period, we find that both the forward-looking content and the net positivity of CEO disclosures decline in CEO tenure. We show that this relation is specific to CEOs, since non-CEO executives’ disclosure patterns in conference calls are unrelated to their CEO’s tenure. Further, we find that externally hired and inexperienced CEOs are more future-oriented, and younger CEOs exhibit greater optimism in their disclosures. The negative relation between disclosure styles and CEO tenure is observed for CEOs appointed following both orderly and forced turnovers. Our findings are robust to alternative empirical choices regarding sampling, specifications, and measures.

Our results are consistent with new CEOs providing more uncertainty-resolving information to satisfy increased outsider demand for information caused by the heightened uncertainty regarding their managerial ability to enhance firm value. Since the uncertainty about their ability also exacerbates CEOs’ career concerns, newly appointed CEOs use their disclosures to

³⁴ “Panel datasets are more informative, provide greater variability, less collinearity among the variables, greater degrees of freedom, and more efficiency” (Baltagi 2008).

³⁵ These results are consistent with Li et al. (2014), who examine CEOs’ personal attributes cross-sectionally and also find a negative association between CEO tenure and the amount of disclosures. Li et al. (2014, Table 5) report the standardized coefficient of -0.14 for CEO tenure (i.e., the number of standard deviation changes in CEO text for a one-standard-deviation change in CEO tenure). In Table 14, this standardized coefficient for CEO tenure is equal to $(-1.002 \times 0.99)/6.59 = 0.151$, indicating magnitudes similar in economic effects to Li et al. (2014).

TABLE 13
CEO Tenure and CEO Disclosure Style
Presentation and Questions and Answers Sections of Earnings Call

	<i>CEOFLS</i>		<i>CEOTone</i>	
	<i>PT</i>	<i>Q&A</i>	<i>PT</i>	<i>Q&A</i>
<i>CEOTenure</i>	−1.285*** (−3.43)	−0.678* (−1.96)	−0.096** (−1.99)	−0.114*** (−3.43)
<i>UE</i>	−24.072 (−0.85)	−35.972 (−1.04)	12.768*** (3.20)	7.599*** (2.63)
<i>Loss</i>	0.160 (0.34)	−0.182 (−0.40)	−0.067 (−1.20)	−0.017 (−0.39)
<i>Loss × UE</i>	28.349 (0.83)	59.000 (1.42)	−6.531 (−1.49)	−2.200 (−0.56)
<i>ROA</i>	5.437 (0.97)	−11.416* (−1.76)	0.828 (1.17)	0.141 (0.28)
<i>Return</i>	−0.860** (−2.03)	−0.527 (−1.44)	0.314*** (6.56)	0.080** (2.51)
<i>Size</i>	1.870*** (2.93)	0.337 (0.56)	−0.351*** (−3.98)	0.015 (0.26)
<i>MTB</i>	−0.460 (−1.25)	−0.237 (−0.61)	0.078 (1.51)	−0.086** (−2.17)
<i>Leverage</i>	−1.432 (−0.62)	0.562 (0.34)	0.141 (0.48)	0.234 (1.05)
<i>EarnVol</i>	9.745 (1.06)	4.708 (0.49)	1.512 (0.73)	−0.168 (−0.23)
<i>RetVol</i>	10.558 (0.57)	34.303* (1.92)	−7.794*** (−3.46)	−2.398 (−1.46)
<i>SpecialItems</i>	−0.090 (−0.24)	−0.175 (−0.40)	0.060 (1.35)	0.113*** (3.32)
<i>DiscOperations</i>	1.865 (0.48)	−1.214 (−0.28)	0.766 (1.29)	0.959* (1.72)
<i>ΔCAPX</i>	2.108 (0.39)	−1.451 (−0.28)	−1.202* (−1.78)	−0.392 (−0.98)
<i>ΔR&D</i>	−9.455 (−1.46)	5.143 (0.75)	1.486*** (2.74)	0.551 (0.73)
<i>ΔADV</i>	−0.186 (−0.06)	3.938 (1.14)	1.002*** (3.98)	0.320 (1.47)
<i>Following</i>	−0.273 (−1.31)	0.263 (1.26)	−0.008 (−0.33)	−0.016 (−0.85)
Firm-CEO Fixed Effects	Yes	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes	Yes
Observations	5,581	5,581	5,581	5,581
Adjusted R ²	0.360	0.272	0.532	0.377

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from panel regressions of CEO-specific forward-looking disclosures (*CEOFLS*) and CEO-specific tone (*CEOTone*) in the presentation (*PT*) and questions and answers (*Q&A*) sections of the earnings conference call on CEO tenure, and firm characteristics. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of firms with CEOs who were appointed after 2005 is used for the estimation. t-statistics are in parentheses.

All variables are defined in Appendix B.

portray their performance and prospects in a more positive light to favorably influence outsiders' beliefs about their managerial ability. Our study provides large-sample evidence on the changes in CEOs' voluntary disclosure styles over their tenure. While prior research that examines CEO characteristics and firm disclosure policy is largely cross-sectional in nature, our work identifies an important dynamic aspect to this relation. Our evidence is consistent with the identified dynamic being driven by the increased uncertainty driving demand for and supply of information and exacerbating managerial career concerns.

TABLE 14
CEO Tenure and Disclosure Style
Sample of All CEOs with Available Earnings Call Transcripts

	<i>FLS</i>		<i>Tone</i>	
	<i>CEOFLS</i>	<i>NonCEOFLS</i>	<i>CEOTone</i>	<i>NonCEOTone</i>
<i>CEOTenure</i>	−1.002*** (−5.93)	−0.060 (−0.29)	−0.058*** (−2.66)	0.023 (1.04)
<i>UE</i>	−2.862 (−0.24)	5.022 (0.44)	8.777*** (6.35)	9.222*** (7.60)
<i>Loss</i>	0.220 (1.18)	0.510*** (2.70)	−0.048** (−2.44)	−0.124*** (−6.72)
<i>Loss × UE</i>	−6.479 (−0.44)	−8.120 (−0.55)	−3.778** (−2.26)	−5.212*** (−3.48)
<i>ROA</i>	−1.624 (−0.65)	0.589 (0.21)	0.564** (2.06)	1.676*** (7.16)
<i>Return</i>	−0.539*** (−3.46)	0.003 (0.02)	0.198*** (11.09)	0.169*** (10.35)
<i>Size</i>	0.404* (1.87)	−0.430* (−1.73)	−0.069** (−2.55)	−0.048* (−1.81)
<i>MTB</i>	−0.374*** (−2.77)	0.102 (0.62)	−0.011 (−0.71)	−0.004 (−0.24)
<i>Leverage</i>	0.304 (0.35)	−2.689** (−2.29)	−0.005 (−0.05)	0.074 (0.67)
<i>EarnVol</i>	0.024 (0.00)	3.194 (0.61)	0.528 (0.85)	1.068** (2.06)
<i>RetVol</i>	6.031 (0.89)	−6.145 (−0.87)	−3.469*** (−4.56)	−3.996*** (−5.80)
<i>SpecialItems</i>	−0.100 (−0.63)	−0.291 (−1.54)	0.040** (2.48)	0.046*** (3.00)
<i>DiscOperations</i>	−1.035 (−0.47)	1.838 (0.73)	0.402 (1.64)	0.522** (2.08)
<i>ΔCAPX</i>	3.167* (1.72)	−1.526 (−0.87)	0.132 (0.69)	0.350** (2.28)
<i>ΔR&D</i>	3.313** (2.06)	−0.711 (−0.28)	0.289* (1.80)	0.106 (0.55)
<i>ΔADV</i>	4.814** (2.25)	6.625* (1.93)	0.369 (1.01)	0.160 (0.46)
<i>Following</i>	−0.118 (−1.41)	−0.225** (−2.36)	−0.037*** (−4.36)	−0.018** (−2.10)
Firm-CEO Fixed Effects	Yes	Yes	Yes	Yes
Year-Qtr. Fixed Effects	Yes	Yes	Yes	Yes
Observations	18,660	18,660	18,660	18,660
Adjusted R ²	0.382	0.349	0.565	0.539

*, **, *** Denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

This table shows the estimated coefficients from OLS regressions of the amount of CEO-specific forward-looking disclosures (*CEOFLS*), amount of forward-looking disclosures by other executives (*NonCEOFLS*), CEO-specific tone (*CEOTone*), and other executives' tone (*NonCEOTone*) in earnings conference calls on CEO tenure and firm characteristics. Firm-CEO fixed effects, year-quarter fixed effects, and the constant are included in each regression, but are not reported. Reported statistics are based on robust standard errors and clustering at the firm level. A sample of all firms with available earnings call transcripts where a CEO participates in the call is used for the estimation. All variables are defined in Appendix B.

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APPENDIX A

Examples of CEO Speech in Earnings Conference Calls

This appendix provides two examples of CEO disclosures in earnings calls. We manually classify CEO statements into three categories: (a) statement most likely referring to the current fiscal quarter (highlighted in bold), and (b) statements related to the future (highlighted in italic). All other statements are in black. Full-text transcripts are available at: www.seekingalpha.com/.

Example 1. Dominion Resources q3 2010 Earnings Call, CEO (Thomas Farrell) Introductory Remarks

Good morning. Before getting started, please note that we posted a supplemental information kit on our website this morning, and I will refer to several slides in the kit during our discussion. The kit is intended to provide reference material to help you understand and model the new Dominion.

For the past nine months, we have conducted a review of Dominion's present [sic] business strategy. We considered market data, had multiple conversations with major institutional shareholders, consulted with various advisers, and had extensive internal discussions and debate. We have completed that process and reviewed our preferred approach with our board.

We have concluded that pursuing the sale of all of our E&P reserves, with the exception of those located in Appalachia, would result in the right mix of assets for Dominion in the long-term. By redeploying the net cash proceeds of any sale into debt reduction, stock buy-backs, and expansion of our remaining businesses, we believe shareholders would benefit from solid, reliable growth from a complementary set of assets that are among the best in their industry groups.

The new Dominion would be refocused and stronger, and would continue to stand as one of the premier integrated electric and gas utilities available to investors.

Dominion's new profile would be a company with:

- *One of the strongest electric utility franchise areas in the country;*
- *One of the premier gas LDCs in the country, East Ohio Gas;*
- *A successful and growing retail program;*
- *One of the industry's best nuclear operations;*
- *A successful and diverse merchant electric generation portfolio;*
- *One of the nation's most active L&G facilities, with plenty of room to expand;*
- *One of the world's largest natural gas storage systems;*
- *A natural gas pipeline business that ranks among the most efficient and profitable in the industry; and*
- *A geographically advantaged and low-risk oil and gas operation, with almost 9,000 producing wells and 7,000 prospective drilling sites.*

These businesses, on a consolidated basis, should grow operating earnings per share at a rate of 4 percent to 6 percent annually. A breakdown of this growth profile can be found on slide 9 of the kit.

Dominion's risk profile would be reduced, as E&P would be a substantially smaller portion of our consolidated earnings. We project our Appalachian production to contribute less than 5 percent of 2008 operating earnings post-sale, about the same percentage it represented before our merger with CNG in the year 2000.

The sensitivity of Dominion's earnings to movements in the price of oil and gas would be approximately two-thirds lower after a sale. Moreover, the remaining commodity exposure would be more predictable and easier to hedge going forward.

Our present business mix has made it difficult for our traditional shareholders to understand Dominion. Despite our best efforts, we have not been able to convince the investment community that the combination of our high quality E&P reserves and the results delivered by our management team and employees, deserve the same earnings credit as more highly-valued, utility type assets.

The percentage of our earnings derived from our E&P business exceeds the comfort level of those investing in the utility sector. This is reflected in Dominion's significant discount to its utility peer group.

Our E&P management team and employees have created a great deal of value for our shareholders, but we do not believe it will ever fully be recognized in Dominion's stock price. At the same time, we expect the premier quality of our reserves, operations, and employee group to produce significant cash proceeds.

After-tax cash would first be used to reduce debt by an amount that would be necessary to maintain our existing credit metric targets. We believe this, combined with a lower business risk profile, will enable us to maintain or improve our credit ratings at both Dominion and CNG.

We would use the remainder to reduce our share count through stock buy-backs, unless specific investments are available in our remaining businesses that would provide a superior alternative. We expect, and have been advised, that once any sale is completed, our per share earnings times an appropriate industry multiple would result in a higher stock value than if we continued on our entire E&P portfolio.

I want to make it clear that our strategic review process is complete and that we intend to keep the resulting business profile for the foreseeable future.

*However, our review of individual assets in each business unit will continue unabated. We will work to improve the return on invested capital across the company. You will appreciate that this has not been a precipitous or an easy decision. **We are blessed with one of the nation's best oil and gas operations and outstanding employees.***

It is our belief that not only will Dominion shareholders benefit, but our E&P employees will have more opportunity to apply their skills and talents in a company grounded in the oil and gas industry.

*Some of you may wonder why we would retain our Appalachian reserves and production. There are a number of factors that support this decision. **As you can see, on our asset map shown on slide 8 of the information kit, these reserves are located in the middle of our gas pipeline, gathering, and storage businesses. Oil and gas production activities help us***

maximize the value of our entire investment in the region. Drilling costs per well are relatively low, and we have experienced a 98 percent success rate in the region over a long period of time.

The location of this production creates a positive basis differential compared to other producing areas. Annual depletion rates are low, which reduces capital needed to replace lost production. Finally, retaining over 1 trillion cubic feet of reserves provides us an indirect hedge for the financial impact of potential CO2 regulations.

There is a lot of industry interest in all of our E&P assets. We have already fielded several reverse inquiries, even before this announcement today.

The sale of all the reserves to one buyer is our preferred approach for a number of reasons. We believe we will receive a premium from a strategic buyer, and our employees deserve a chance to keep this program together, as long as we obtain maximum value.

We will begin a formal sales process in mid-February, after our 2006 reserve audit is complete. Our expectation is that any transaction from this process would close around mid-2007.

During our strategic review, we thoroughly analyzed an IPO spin-off and determined that it would not create shareholder value equal to the anticipated value which we expect to be created by a sale. We will, however, revisit this option in the event that we do not achieve expected value for our sales process.

For Dominion, today's announcement is not about retrenchment. *It is about refocused growth. We are looking at significant earnings expansion over the next few years from a variety of factors we have reviewed previously. Also, we have our sights set on other opportunities.*

There are many positive events on the horizon. For example, starting next July, Dominion Virginia Power will revert to a traditional fuel clause. We will not be eligible for a one-time true-up for periods prior to July of 2007. We will, however, be eligible for true-ups in all future periods. Had we functioned under a traditional fuel clause this year, it would have net more than an expected \$300 million in additional operating income for 2006.

We also will benefit next year from higher prices, realized for our New England generating units, and capacity payments under forward-capacity markets, as well as continued growth in our electric distribution service area. We are adding about 40,000 new customer accounts per year.

You are also aware of the discussions about building a new coal fire power station in Southwest Virginia. We are in the process of obtaining an early site burning to build a third nuclear unit at North Anna Power Station if and when the right regulatory and financial incentives materialize.

The Cove Point LNG expansion is scheduled to come online in 2008.

We are also looking for other growth opportunities in our remaining core business lines. We will seek opportunities to produce strong returns on invested capital and to create long-term shareholder value. With our strategy more clear and a financially stronger company, we will be positioned to pursue those opportunities.

Today's announcement is a result of a new way of looking at our business. Earnings and cash flow are important, but so is maximizing the risk reward ratio of our investments.

Although Dominion pays a sizable dividend from the income we derived from our existing operations, we believe that we can pay out a higher percentage for the remaining businesses, which will have a lower risk and more consistent overall earnings profile than we have today.

It is too early to change our dividend policy, but management will address the possibility of an enhanced quarterly dividend following any transactions related to our strategic decision.

We have been ably assisted in this work by JP Morgan, Lehman Brothers, and Juniper Advisory LP. Advising us on market perception of our existing and future activities has been Steve Fleishman of Merrill Lynch. Our legal advisers are Baker Botts and McGuire Woods.

There is much work to do to accomplish the sale of these E&P properties. Our whole organization has geared up to make it a successful process. Once again, as I have done on many previous occasions, let me say how much we appreciate the professionalism and excellence demonstrated by Duane Radtke and the entire Dominion E&P team.

I am now going to turn the call back over to Tom Chewing to discuss some of the financial aspects of the transaction.

Example 2. Dominion Resources Q3 2016 Earnings Call, CEO (Thomas Farrell) Introductory Remarks

Good morning, everyone, and thank you for joining us. **Our operating and safety performance continued to be strong during the third quarter. We finished the quarter again as the top-rated company in the Southeast in employee safety performance. The fossil and hydro utility fleet achieved a year-to-date equivalent forced outage rate of 2.7 percent through September, which is its best on record. Our nuclear fleet achieved a third quarter capacity factor of 97 percent, excluding refueling outages. Our Kewaunee Nuclear power station in Wisconsin won the American Nuclear Society 2010 Utility Achievement Award for Outstanding Performance in Plant Operations. Kewaunee was recognized**

for demonstrating outstanding performance in operations as the top-performing plant and as the plant showing the most improved and sustainable performance in overall operations.

We are pleased with our 2011 to 2015 regulated infrastructure growth plan we laid out in early September. Our five-year plan identifies over \$2 billion in growth capital investments from across all of our regulated businesses. Our plan continues to develop on schedule and on budget. The Bear Garden combined cycle plant was 85 percent complete at the end of the third quarter. All major equipment is in place and construction continues per plan, with about 550 workers on site. Assembly of the gas turbine is complete with commissioning progressing to support a first fire later this year. *The 580-megawatt combined cycle plant is scheduled for commercial operation in the spring of next year.*

Virginia City Hybrid Energy Center was 74 percent complete at the end of the third quarter. The 585-megawatt project is proceeding according to plan with over 2,000 workers on site. The steam turbine generator and the main transformer were delivered and placed under foundations, which represents the last of the major engineered equipment deliveries. *The plant is scheduled for commercial operation in 2012. We have had no delays or cost overrun on either of these projects, which in total represent over \$2.4 billion of capital investment.*

We continue to make significant progress on our Warren County project, a three-on-one combined cycle facility, with projected capacity of up to 1,300 megawatts. The local county has approved the revised conditional use permit. Dominion held its required air permit informational briefing in May, and the draft air permit was issued for public comment earlier this month. We expect the final air permit in December. We have completed the steam turbine and gas turbine selection process, as well as executed the natural gas transportation receive agreement to supply fuel to the power station. *We plan to file for approval of this plant with the Virginia State Corporation Commission early next year.*

With respect to North Anna 3, we have concluded that the capital required for other new generation, pending environmental compliance and electric transmission upgrades, requires us to slow development of the third reactor at that facility. *Dominion will continue to work toward obtaining a combined operating license for North Anna Unit 3 incorporating the US-APWR design by Mitsubishi Heavy Industries. We and MHI will also continue our engineering and preliminary site development work. We intend to reassess the schedule for construction of the unit as we approach the issuance of the COL, which we expect to obtain in early 2013. We look forward to continuing work on this important project for Virginia. Our \$10.2 billion five-year growth plan is unaffected by this decision.*

Our two major electric transmission projects are also on schedule to be fully in-service by next summer. Two phases of the 500kV Meadow Brook-to-Loudoun line are complete and in operation, while work continues on the remaining phase. Work on the 500kV Carson-to-Suffolk line is about 50 percent complete. Yesterday, we energized our Pleasant View-to-Hamilton 230kV line in Northern Virginia. The project was on time and on budget. Also, yesterday, PJM announced its intention to recommend to the PJM Board the reconductoring of our Mt. Storm-to-Doubs line as part of the 2010 Regional Transmission Expansion Plan. We anticipate formal PJM approval in December, in line with our expectations since we first announced this project at our May financial conference.

Dominion Energy continues to take advantage of its strategic location in the Marcellus Shale region. For example, recently, DTI signed a new 15-year firm transportation agreement to move 100,000 dekatherms per day of Marcellus Shale volumes. This is our third Marcellus-related project announcement in as many months. The Marcellus 404 project also continues to move forward. We have been meeting with producers to gauge interest and have further investigated the front-end engineering and design. Our two major energy construction projects, Gathering Enhancement and the Cove Point Pier Expansion, remain on schedule and on budget.

Now a few minutes on the progress in our regulatory proceedings. In September, parties to the base rate case in North Carolina filed a settlement agreement with the North Carolina Utilities Commission. The stipulating parties agreed that non-fuel based revenues will increase by \$7.7 million, our first base rate increase in North Carolina since 1993. Also, a significantly greater percentage of future purchased power expenses will now be recovered through the fuel clause rather than base rates, reducing risk for the company. *The settlement, if approved by the Commission, would resolve all outstanding issues in the current base and fuel cases in North Carolina. We expect the decision in that proceeding before the end of the year. It is worth noting that our customer rates in both Virginia and North Carolina are well below the averages for the region, as well as the nation as a whole.*

On June 30, we filed our annual update to the riders for the Virginia City and Bear Garden Power Stations. These filings cover the financing costs for both projects for 2011, as well as the operating and maintenance costs for Bear Garden once it goes into commercial operation in the spring of next year. *If approved by the State Corporation Commission, the change in rates will be effective on April 1.*

I want to cover a few upcoming events for the fourth quarter. *We expect to complete a 60-megawatt upgrade at our Surry Nuclear Power Station and to have a successful first firing at the Bear Garden Power Station. We expect a final air permit for our Warren County project in December. We also expect PJM approval of the Mt. Storm-Doubs line in December. On the regulatory front, we look forward to the North Carolina Commission's consideration of the settlement agreement. Finally, we*

will submit for approval, by our Board of Directors, our recommendation of a 60 percent to 65 percent target dividend payout ratio, including a 7.7 percent increase in the dividend rate for 2011 over 2010.

Overall, we are pleased with results for the third quarter. *Our infrastructure growth program continues to provide the foundation for sustained earnings growth in the future.*

Thank you, and we are now ready for your questions.

APPENDIX B

Variable Definitions and Data Sources

Variable	Definition	Source
<i>CEOFLS</i>	The number of words in CEOs' forward-looking statements in the conference call, divided by the total number of CEO words in the conference call.	Conference calls are from: https://seekingalpha.com
<i>CEOFLSAlt</i>	The number of words in CEOs' forward-looking statements in the conference call, divided by the total number of words in the conference call excluding analyst-specific text.	Conference calls are from: https://seekingalpha.com
<i>CEOTone</i>	The number of positive words minus the number of negative words from Loughran and McDonald's (2011) dictionary that occur in CEO statements in the conference call, divided by the total number of CEO words in the conference call.	Conference calls are from: https://seekingalpha.com
<i>NonCEOFLS</i>	The number of words in forward-looking statements of non-CEO officers in the conference call, divided by the total number of these officers' words in the conference call.	Conference calls are from: https://seekingalpha.com
<i>NonCEOTone</i>	The number of positive words minus the number of negative words from Loughran and McDonald's (2011) dictionary that occur in statements of non-CEO officers in the conference call, divided by the total number of these officers' words in the conference call.	Conference calls are from: https://seekingalpha.com
<i>CEOWords</i>	The natural logarithm of the number of words spoken by a CEO in the conference call.	Conference calls are from: https://seekingalpha.com
<i>AllWords</i>	The natural logarithm of the number of words spoken by all company executives (including CEO) in the conference call, excluding analyst-specific text.	Conference calls are from: https://seekingalpha.com
<i>CEOTenure</i>	The natural logarithm of CEO tenure (in quarters), where CEO tenure is the quarterly difference between current-period end date and CEO appointment date.	Execucomp
<i>CEOAge</i>	The natural logarithm of CEO age.	Execucomp
<i>ExtHire</i>	Indicator variable that equals 1 if the CEO was not employed by the company before taking this CEO position, and 0 otherwise.	BoardEx
<i>NoExecExp</i>	Indicator variable that equals 1 if the CEO was neither a CEO nor a CFO at another company before taking this CEO position, and 0 otherwise.	BoardEx
<i>YoungCEO</i>	Indicator variable that equals 1 if the CEO was under the age of 52 (median CEO age in our sample) when appointed, and 0 otherwise.	Execucomp
<i>UE</i>	Actual earnings per share (EPS) minus analyst consensus forecast of one- or two-quarters-ahead earnings issued or reviewed in the last 60 days before earnings announcement, divided by stock price at the end of quarter, winsorized at 1 percent and 99 percent.	I/B/E/S
<i>Loss</i>	Indicator variable that equals 1 if actual EPS is less than zero.	I/B/E/S
<i>ROA</i>	Earnings before extraordinary items scaled by total assets, winsorized at 1 percent and 99 percent.	Compustat
<i>Return</i>	Annual buy-and-hold return over the past year, calculated using monthly data.	CRSP
<i>Size</i>	The natural logarithm of the market value of equity at the end of the previous quarter.	Compustat
<i>MTB</i>	Market value of equity plus book value of liabilities divided by book value of assets, measured at the end of the previous quarter, winsorized at 1 percent and 99 percent.	Compustat
<i>Leverage</i>	Long-term debt-to-total assets ratio at the end of the previous quarter, winsorized at 1 percent and 99 percent.	Compustat
<i>RetVol</i>	The volatility of the residual return from the Fama-French three-factor model over the three-month period prior to the earnings announcement.	CRSP
<i>EarnVol</i>	Standard deviation of earnings, calculated using earnings scaled by total assets in the last 20 quarters, with a minimum of eight quarters required.	Compustat
Δ CAPX	Change in capital expenditures, scaled by total assets, in year $t-1$ relative to year $t-2$, winsorized at 1 percent and 99 percent.	Compustat
Δ R&D	Change (in year $t-1$ relative to year $t-2$) in the research and development (R&D) expense, scaled by total assets, with missing values in R&D expense treated as zero, winsorized at 1 percent and 99 percent.	Compustat
Δ ADV	Change (in year $t-1$ relative to year $t-2$) in advertising expense, scaled by total assets, with missing values in advertising expense treated as zero, winsorized at 1 percent and 99 percent.	Compustat
<i>DiscOperations</i>	Gain (loss) from discontinued operations scaled by total assets, winsorized at 1 percent and 99 percent.	Compustat
<i>Following</i>	The natural logarithm of the number of analysts that issue an earnings forecast for a given firm.	I/B/E/S

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